UCCN 2003
UCCN 2243

TCP/IP Internetworking, Internetwork Principles & Practices

(Lecture 02)

IPv4 Subnetting & Supernetting

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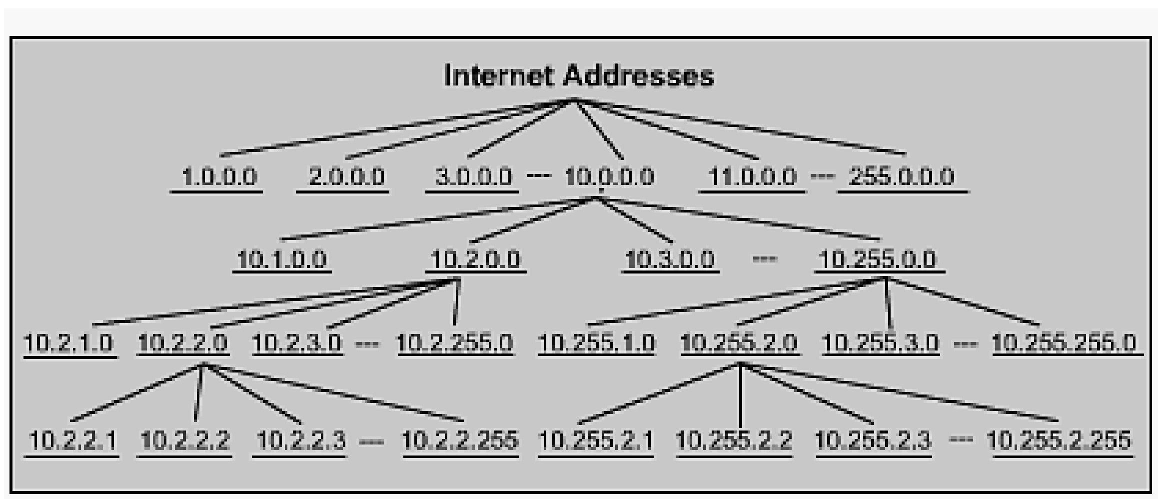
UCCN1003: IP Subnet Rules #14

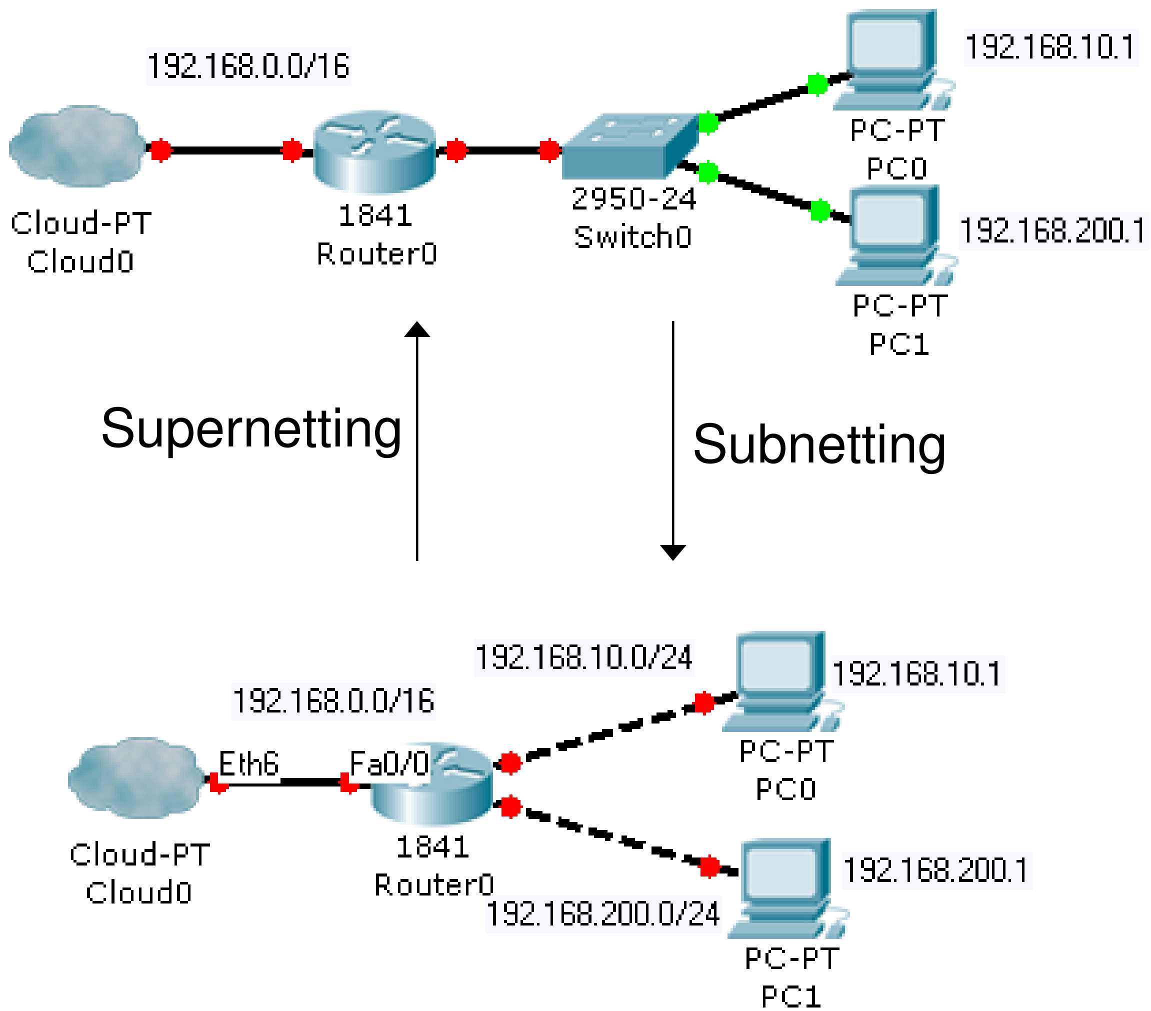
- The principle of IP design in subnetting and supernetting
- Subnetting
  - 1 bigger IP subnet split into a few smaller IP subnets.
- Supernetting
  - 2 or more smaller IP subnets are joined into a larger IP subnet.

supernetting



subnetting

UCCN1003: IP Subnet Rules #14



Quick Quiz: Subnet and Supernet

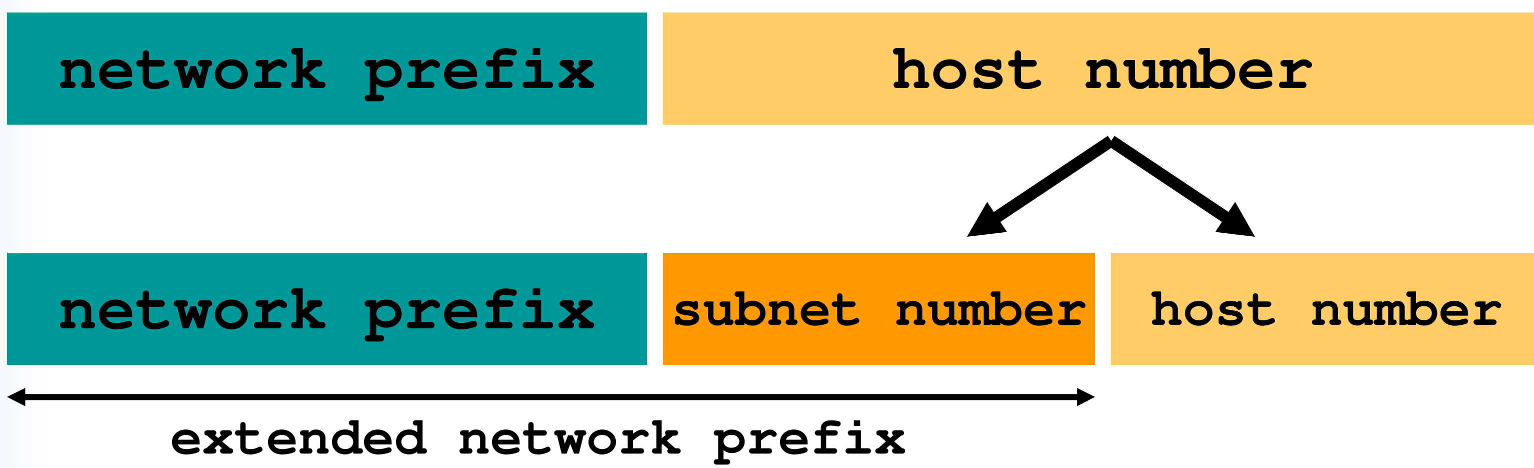
- Subnetting or supernetting will produce more hosts within a subnet?
- Subnetting or supernetting will require more routers?
- Supernetting will require more switch ports or less switch ports?
- Supernetting will result in longer network ID or shorter network ID?

Answer

- Supernetting
- Subnetting
- More switch ports
- Shorter network ID

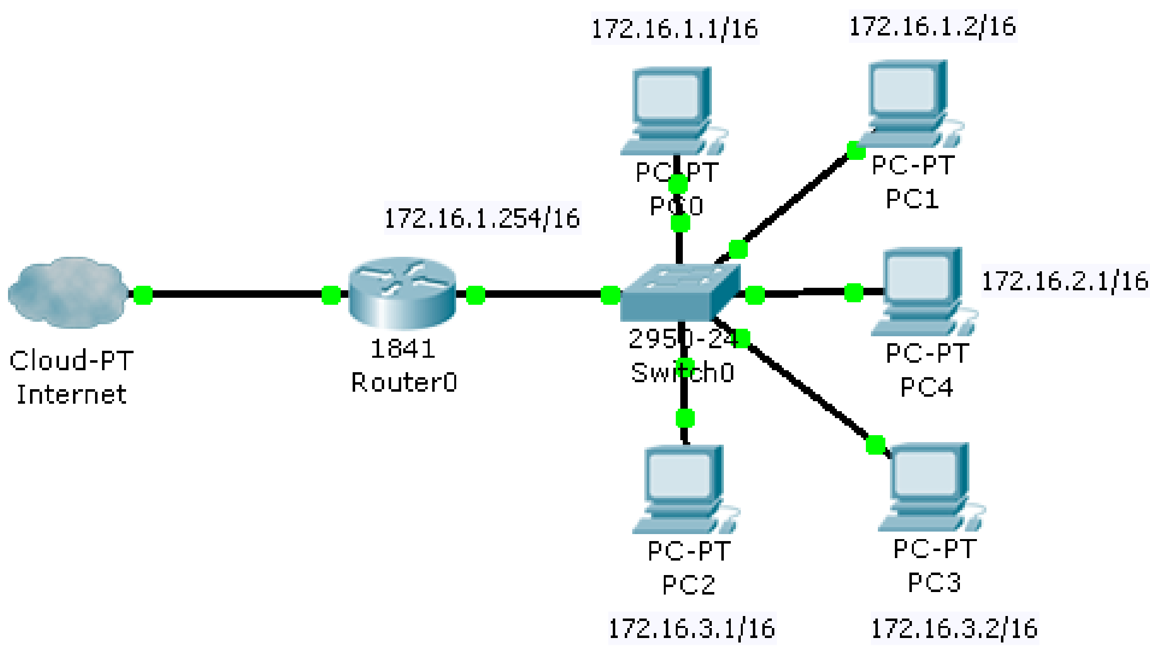
UCCN1003: IP Subnet Rules #14

- Split the host number portion of an IP address into a subnet number and a (smaller) host number.
- Result in a few blocks of IP addresses.
- The subnet mask will be extended with more '1'
- The subnet mask will get "larger". (e.g: /16 => /20)



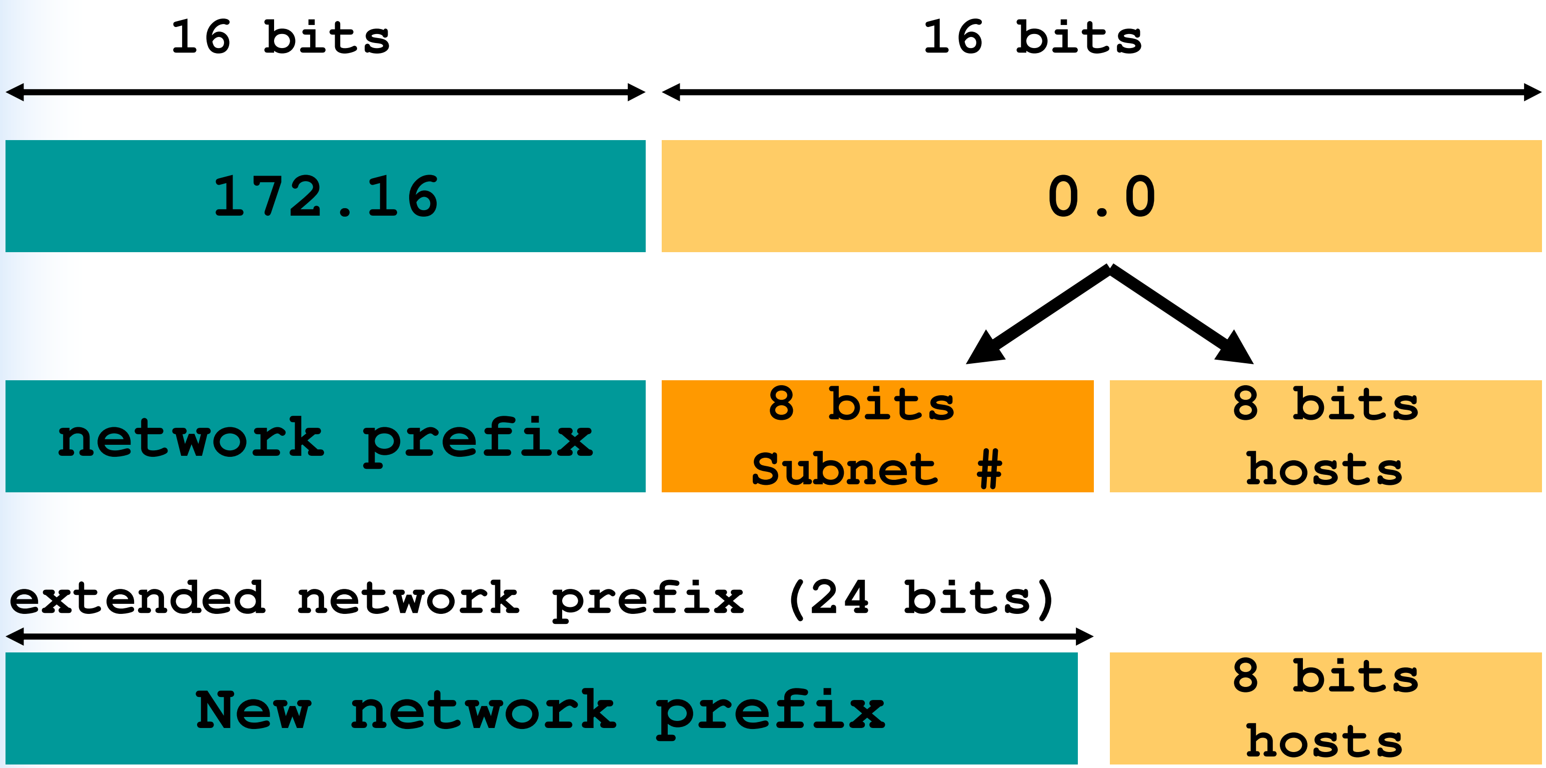
UCCN1003: IP Subnetting example (1)

- Perform subnetting for the following LAN.
  - 172.16.0.0/16
- Specification:
  - Subnet number = 8 bits.



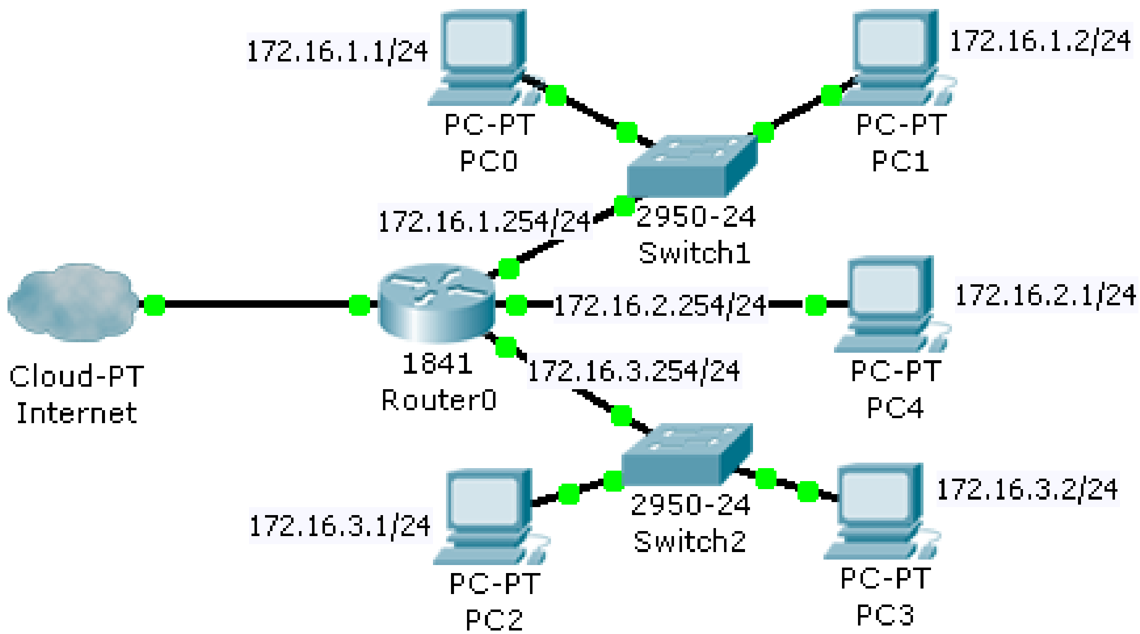
UCCN1003: IP Subnetting example (2)

- Original subnet mask = /16
- New extended subnet mask = /(16 + 8) = /24



UCCN1003: IP Subnetting example (3)

- General Implementation:
  - Subnet 1 network into 3 LAN
  - 1 network ID (172.16.0.0/16) becomes 3 network ID
    - (172.16.1.0/24, 172.16.2.0/24, 172.16.3.0/24)
  - All subnet mask has been changed from /16 to /24
  - 1 gateway -> 3 gateways

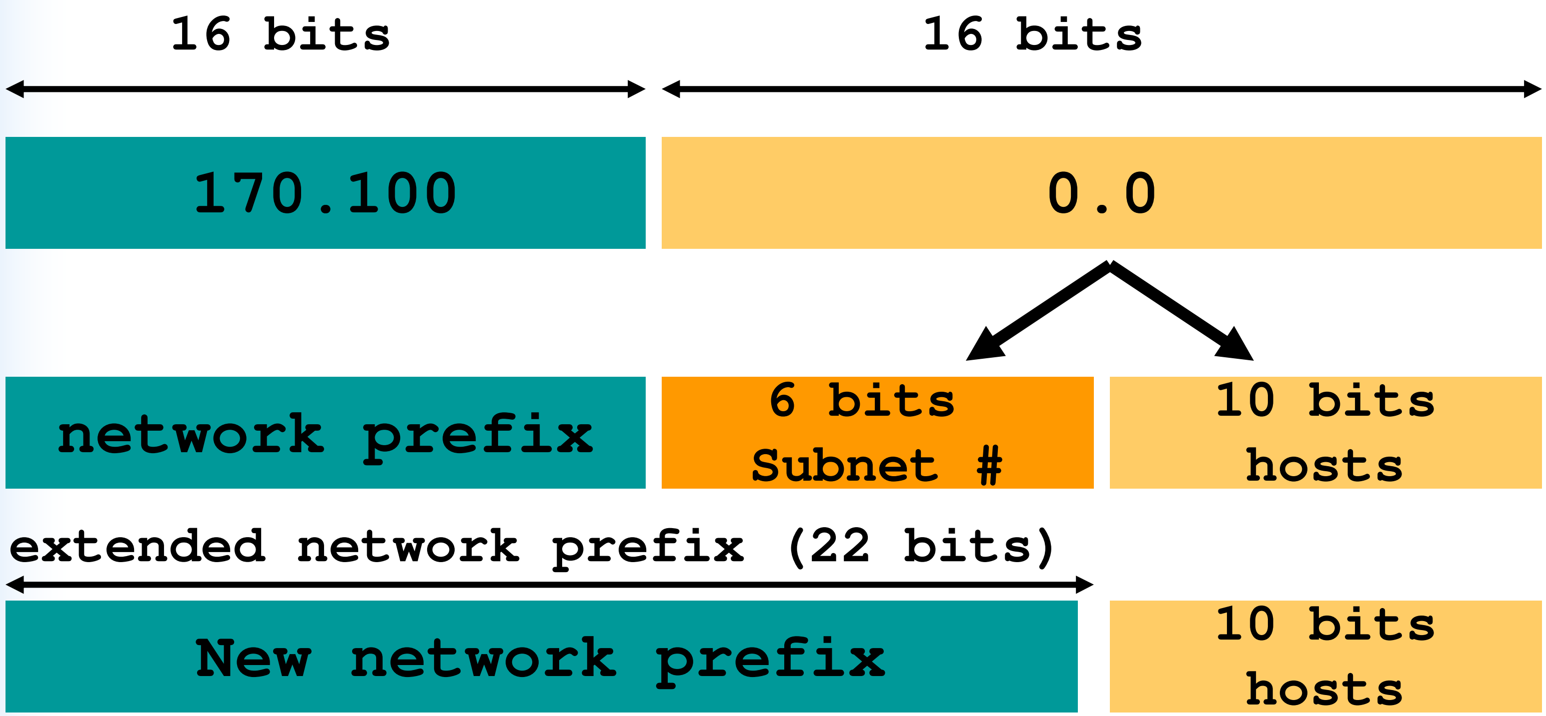


Quick Quiz:

- If you have a network 170.100.0.0/16, the number of subnet bits allocated is 6, then:
  - What will be the new subnet mask?
  - How many more gateway IP addresses do you require?

Answer

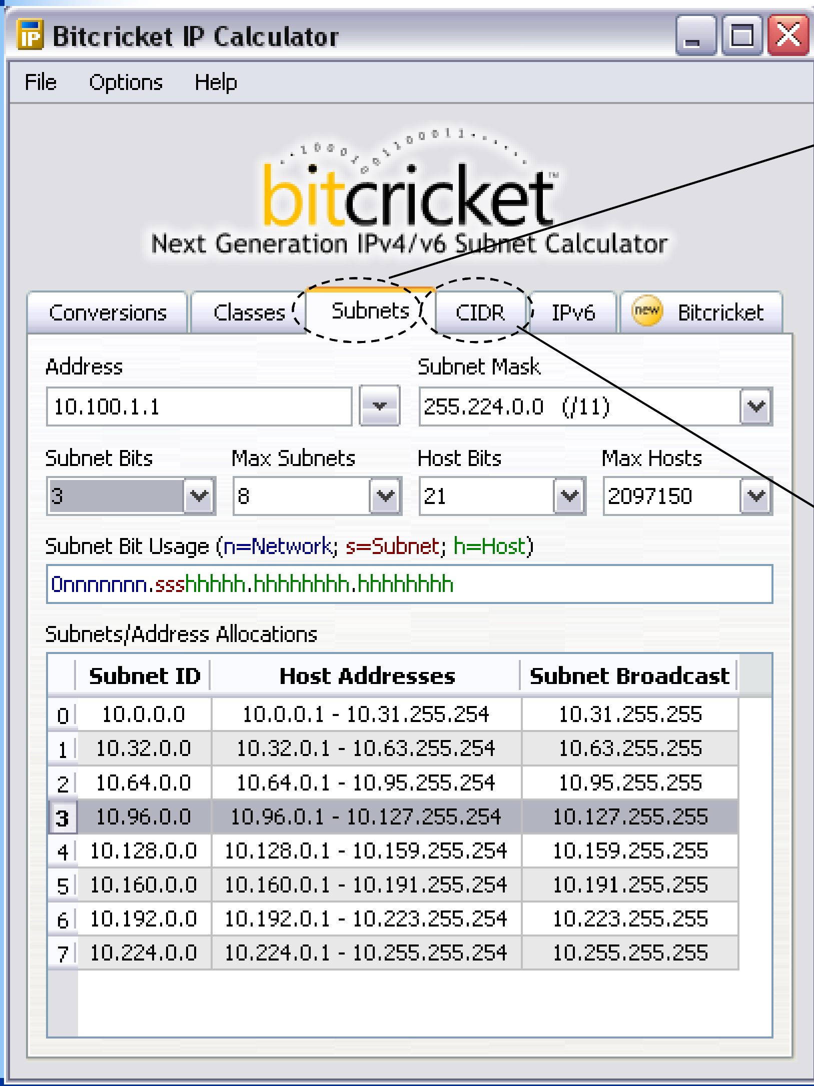
- /16 + 6 = /22 = 255.255.252.0
- Assume that originally you need 1 gateway IP address. 6 subnet bits will require 2^6 gateway addresses = 64
  – Maximum you need (64 – 1) = 63 more gateway IP address.



UCCN2243: Types of Subnetting

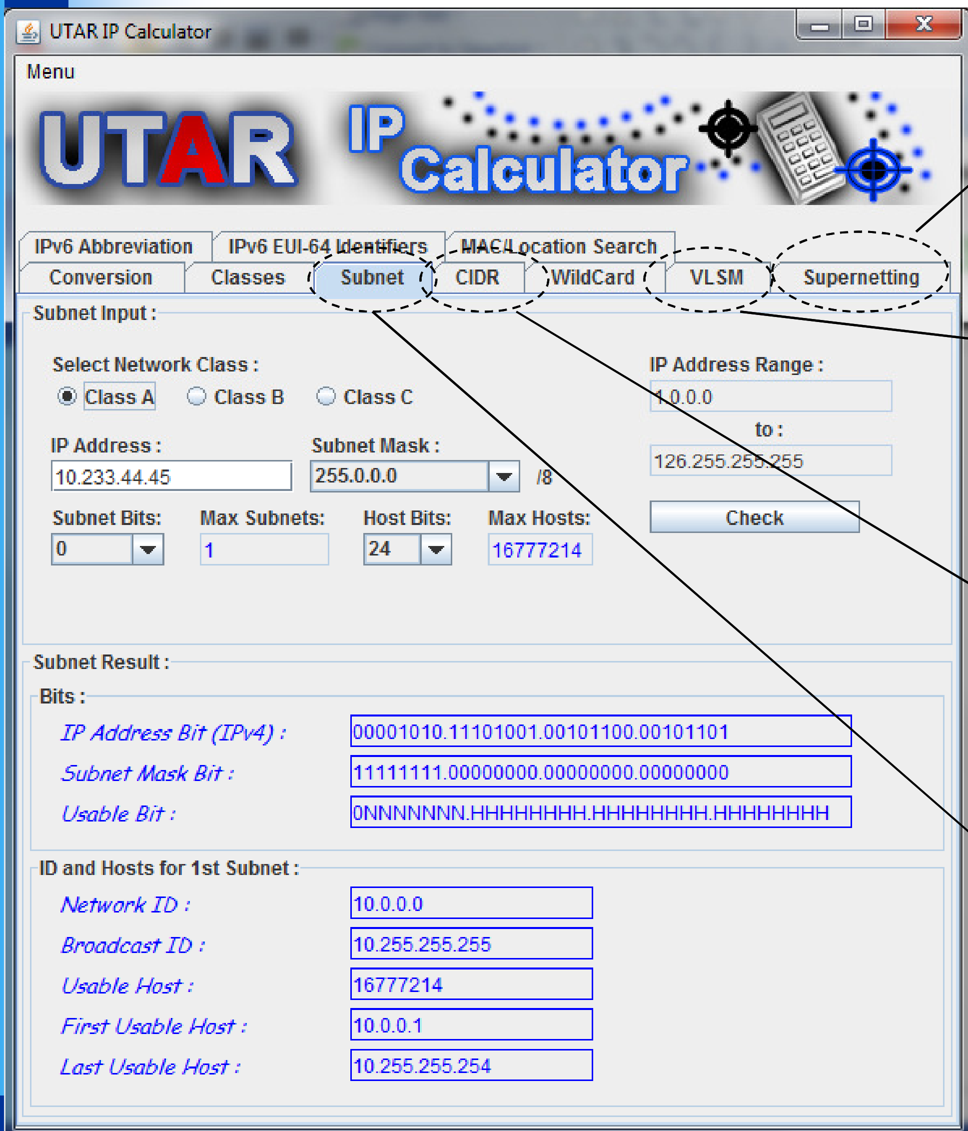
- Traditional classful subnetting
  – Based on the IP class to perform subnetting
  – Design with equal size subnet mask across “the network”
- VLSM (variable subnet mask)
  – Design a network with subnets of different “size” of subnet masks
  – More efficient use of the IP addresses

Using Bitcricket IP Calculator



- Use “Subnet” to perform traditional classful subnetting.
- Use CIDR to help to design the VLSM subnetting.

Improved: UTAR IP Calculator



- Supernetting
- Subnetting using VLSM
- Classless Inter-domain Routing
- Subnetting using traditional classful method.

Traditional Classful Subnetting

Traditional classful Subnetting

- First, determine which class does the IP belongs to
  - Class A (24 host bits), B (16 host bits) or C (8 host bits)
  - For example, IP 10.x.x.x is a class A IP, so you can't treat it as a class B IP.
- Second, determine how many subnet bits you want to allocate.
- Third, compute the new subnet mask.
- Fourth, plan the IP range of each subnet.

Traditional Classful Subnetting

- In this subnetting scheme, you <u>can't</u> use the subnet bits which is all '0's and all '1's
  - There is a way to "overcome" this, will discuss later in the slides.

Original Network bits	Subnet bits = "00.." or "11.."	Host bits

- In fact, you will "lose" a lot of usable host IP addresses in the form of
  - Subnetwork network ID (all host bits = '0')
  - Subnetwork broadcast address (all host bits = '1's)
  - Gateway IP addresses

UCCN2243: IP Subnet Zero (1)

- When you work with classical subnetting, you always have to eliminate the subnets that contain either all zeros or all ones in the subnet portion.
- Hence, you always used the formula $2^N - 2$ to define the number of valid subnets created.
- However, **Cisco devices can use those subnets, as long as the command "ip subnet-zero" is in the configuration.**

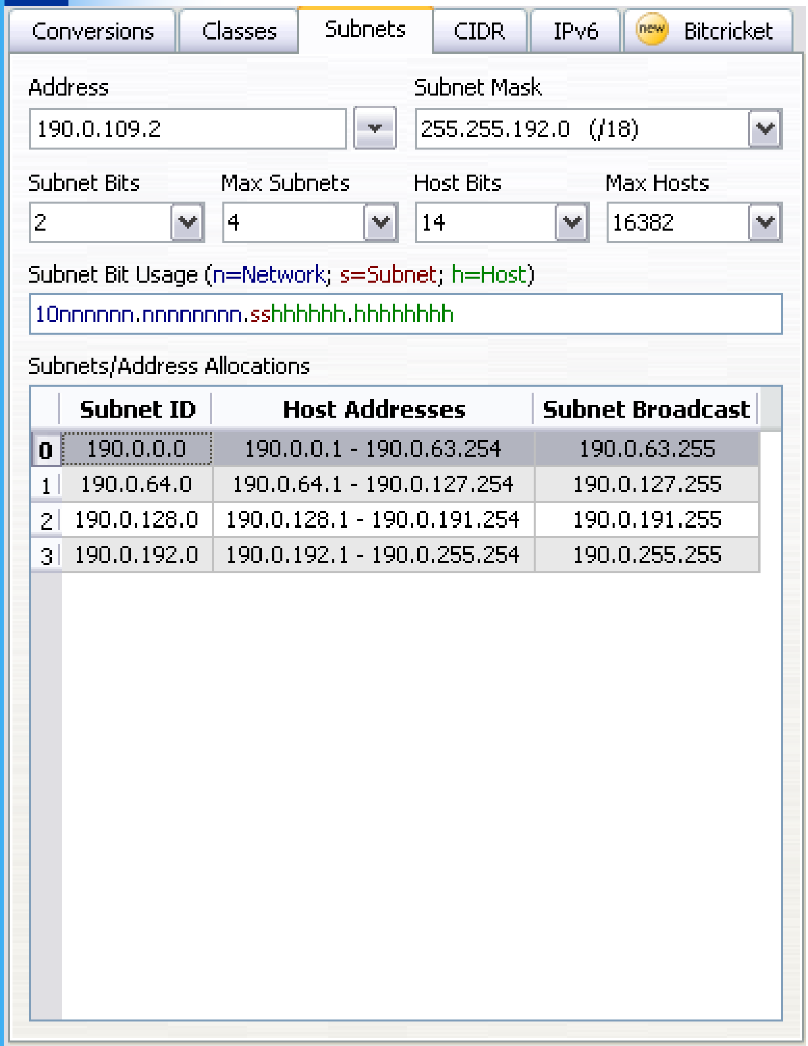
UCCN2243: IP Subnet Zero (2)

- This command **is on by default in Cisco IOS Software Release 12.0 and later**; if it was turned off for some reason, however, you can re-enable it by using the following command:

```
Router(config)#ip subnet-zero
```

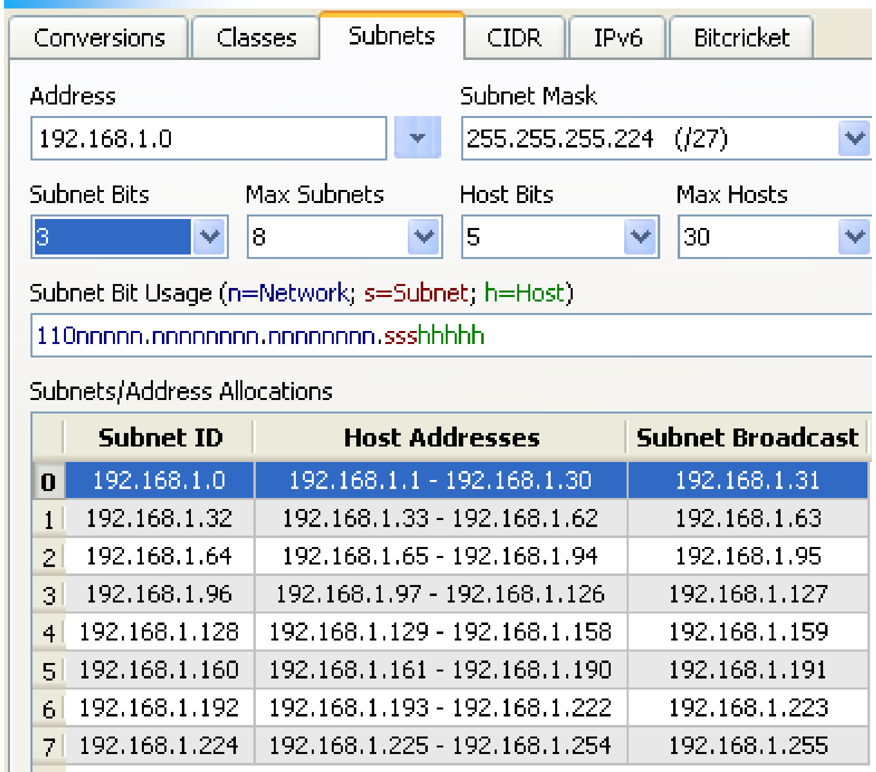
- Now you can use the formula $\mathbf{2^N}$ rather than $\mathbf{2^N - 2}$.

UCCN2243: IP Subnet Zero (3)



- With "ip subnet-zero", we can use the previously two "forbidden" subnets in the traditional classful subnetting.
- In the example, we can now use subnet '0' and subnet '3'.

UCCN2243: IP Subnet Zero (4)



	Subnet ID	Host Addresses	Subnet Broadcast
0	192.168.1.0	192.168.1.1 - 192.168.1.30	192.168.1.31
1	192.168.1.32	192.168.1.33 - 192.168.1.62	192.168.1.63
2	192.168.1.64	192.168.1.65 - 192.168.1.94	192.168.1.95
3	192.168.1.96	192.168.1.97 - 192.168.1.126	192.168.1.127
4	192.168.1.128	192.168.1.129 - 192.168.1.158	192.168.1.159
5	192.168.1.160	192.168.1.161 - 192.168.1.190	192.168.1.191
6	192.168.1.192	192.168.1.193 - 192.168.1.222	192.168.1.223
7	192.168.1.224	192.168.1.225 - 192.168.1.254	192.168.1.255

- With "ip subnet-zero",
  - The 1st subnet is subnet number = 0
  - In the example, 1st subnet will be 192.168.1.1 to 192.168.1.30
- Without "ip subnet-zero"
  - The 1st subnet is subnet number = 1
  - In the example, 1st subnet will be 192.168.1.33 to 192.168.1.62

Important!!

- Currently, both Cisco and Huawei automatically support “IP subnet-zero”.
- From now on, “IP subnet-zero” is assumed to be supported in all routers by default, until stated otherwise.

Traditional Classful Subnetting

- There are two ways to determine the subnet bits (you can choose either one the following):
  - The number of subnets that you want to have in the network that you are going to "split"
    - For example, you want 10 subnets in the network, then you need S subnet bits. ($2^S >= 10$)
  - The maximum number of hosts that you want to allocate
    - For example, if you have a class B IP address, and you need maximum1000 hosts in any one of the subnet, then you need S subnet bits where (32 = 16 + S + H bits), $2^H >= 1000$ => H = 10 bits. Then S = 32 – 16 – 10 = 6 bits.
    - 16 is the original host bits for class B IP address.

Quick Quiz

- If I have a class B address, and I need 11 subnets, what should be my
  - Subnet bits
  - Host bits
  - Max number of host in each subnet
  - New subnet mask?

Answer:

- Subnet bits = 4
- Host bits = 12
- Max # of host = $2^{12} - 2 = 4094$
- New mask = /20

Quick Quiz: Number of subnet bits

- How many departments can you design if your company:
  - has a class C IP address;
  - 1 department = 1 subnet (assume NO ip subnet-zero);
  - max host in any one of the department = 18;

Answer

- Class C IP has 8 bits host
- 18 host requires $2^S > 18 \Rightarrow S = 5$
- $8 - 5 = 3$ (subnet bits)
- $2^3 = 8$
- $8 - 2 = 6$ subnets => 6 departments
- 2 represents subnet bits with all '0's and all '1's.

UCCN2243: Steps to Subnetting (1)

- Case Study:
  - You have a Class C address of 192.168.100.0 /24. You need nine subnets (assume NO ip subnet-zero).
  - What is the IP plan of network numbers, broadcast numbers, and valid host numbers?
  - What is the subnet mask needed for this plan?

UCCN2243: Steps to Subnetting (2)

- Since the subnet mask is /24, we only need to focus on the 4^{th} octet.
  - The 4 octet is the host bits, represented by ‘H’
  - The ‘N’ bits is the new “subNet” bits.

You cannot use N bits, only H bits. Therefore, ignore 192.168.100. These numbers cannot change.

Step 1 Determine how many H bits you need to borrow to create nine valid subnets.

$$2^N - 2 \geq 9$$

N = 4, so you need to borrow 4 H bits and turn them into N bits.

Start with 8 H bits	HHHHHHHH
Borrow 4 bits	NNNNHHHH

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UCCN2243: Steps to Subnetting (3)

Step 2 Determine the first valid subnet in binary.

0001HHHH	Cannot use subnet 0000 because it is invalid. Therefore, you must start with the bit pattern of 0001
0001**0000**	All 0s in host portion = subnetwork number
0001**0001**	First valid host number
.	
.	
.	
0001**1110**	Last valid host number
0001**1111**	All 1s in host portion = broadcast number

UCCN2243: Steps to Subnetting (4)

Step 3 Convert binary to decimal.

00010000 = 16	Subnetwork number
00010001 = 17	First valid host number
.	
.	
.	
00011110 = 30	Last valid host number
00011111 = 31	All 1s in host portion = broadcast number

UCCN2243: Steps to Subnetting (5)

Step 4 Determine the second valid subnet in binary.

0010HHHH	0010 = 2 in binary = second valid subnet
00100000	All 0s in host portion = subnetwork number
00100001	First valid host number
.	
.	
.	
00101110	Last valid host number
00101111	All 1s in host portion = broadcast number

UCCN2243: Steps to Subnetting (6)

Step 5 Convert binary to decimal.

00100000 = 32	Subnetwork number
00100001 = 33	First valid host number
.	
.	
.	
00101110 = 46	Last valid host number
00101111 = 47	All 1s in host portion = broadcast number

UCCN2243: Steps to Subnetting (7)

Step 6 Create an IP plan table.

Valid Subnet	Network Number	Range of Valid Hosts	Broadcast Number
1	16	17–30	31
2	32	33–46	47
3	**48**	**49–62**	**63**

Notice a pattern? Counting by 16.

UCCN2243: Steps to Subnetting (8)

Step 7 Verify the pattern in binary. (The third valid subnet in binary is used here.)

0011HHHH	Third valid subnet
00110000 = **48**	Subnetwork number
00110001 = **49**	First valid host number
.	
.	
.	
00111110 = **62**	Last valid host number
00111111 = **63**	Broadcast number

UCCN2243: Steps to Subnetting (8)

Subnet	Network Address (0000)	Range of Valid Hosts (0001–1110)	Broadcast Address (1111)
0 (0000) invalid	192.168.100.0	192.168.100.1–192.168.100.14	192.168.100.15
1 (0001)	192.168.100.16	192.168.100.17–192.168.100.30	192.168.100.31
2 (0010)	192.168.100.32	192.168.100.33–192.168.100.46	192.168.100.47
3 (0011)	192.168.100.48	192.168.100.49–192.168.100.62	192.168.100.63
4 (0100)	192.168.100.64	192.168.100.65–192.168.100.78	192.168.100.79
5 (0101)	192.168.100.80	192.168.100.81–192.168.100.94	192.168.100.95
6 (0110)	192.168.100.96	192.168.100.97–192.168.100.110	192.168.100.111
7 (0111)	192.168.100.112	192.168.100.113–192.168.100.126	192.168.100.127

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UCCN2243: Steps to Subnetting (9)

8 (1000)	192.168.100.**128**	192.168.100.**129**–192.168.100.**142**	192.168.100.**143**
9 (1001)	192.168.100.**144**	192.168.100.**145**–192.168.100.**158**	192.168.100.**159**
10 (1010)	192.168.100.**160**	192.168.100.**161**–192.168.100.**174**	192.168.100.**175**
11 (1011)	192.168.100.**176**	192.168.100.**177**–192.168.100.**190**	192.168.100.**191**
12 (1100)	192.168.100.**192**	192.168.100.**193**–192.168.100.**206**	192.168.100.**207**
13 (1101)	192.168.100.**208**	192.168.100.**209**–192.168.100.**222**	192.168.100.**223**
14 (1110)	192.168.100.**224**	192.168.100.**225**–192.168.100.**238**	192.168.100.**239**
15 (1111) invalid	192.168.100.**240**	192.168.100.**241**–192.168.100.**254**	192.168.100.**255**
Quick Check	**Always an even number**	**First valid host is always an odd #** **Last valid host is always an even #**	**Always an odd number**

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UCCN2243: Steps to Subnetting (10)

Step 9 Calculate the subnet mask.

The default subnet mask for a Class C network is as follows:

Decimal	Binary
255.255.255.0	11111111.11111111.11111111.00000000

1 = Network or subnetwork bit
0 = Host bit

You borrowed 4 bits; therefore, the new subnet mask is the following:

11111111.11111111.11111111.**1111**0000	255.255.255.**240**

NOTE: You subnet a Class B or a Class A network with exactly the same steps as for a Class C network; the only difference is that you start with more H bits.

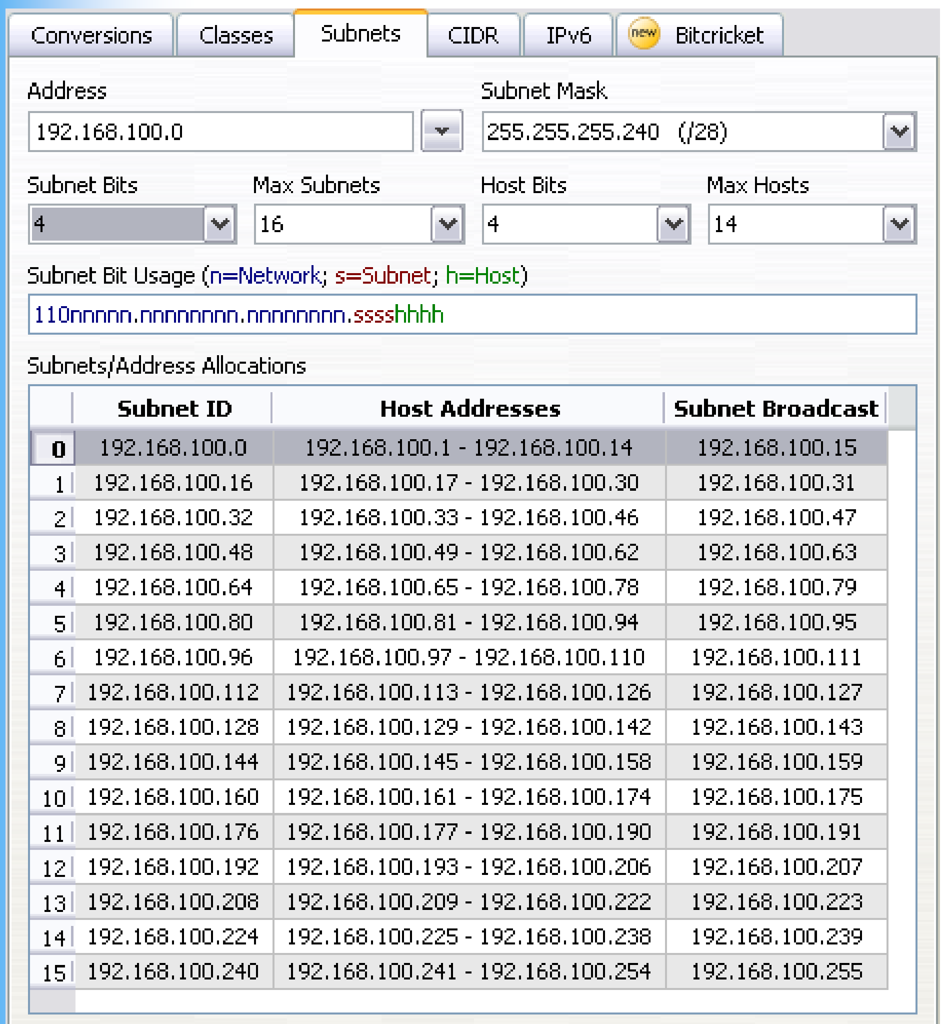
Quick Quiz:

- How many host IP addresses are there in 192.168.100.0/24?
- How many host IP addresses are there in all the “valid” subnets, if 4 bits is used as subnet bits (assume NO ip subnet-zero)?
  - Assume host only consist of end devices PCs, servers, and printers.

Answer:

- $2^8 - 2 = 254$ hosts
- $2^8 - 16 - 16 - 14 - 14 - 14 = 182$
  - 16 = subnet with all ‘0’ subnet bits
  - 16 = subnet with all ‘1’ subnet bits
  - 14 = subnet network address
  - 14 = subnet broadcast address
  - 14 = gateway IP for the subnets
- Compare to the original “supernet”, the subnetting has lost (254-182) = 72 “useful host IP addresses”.

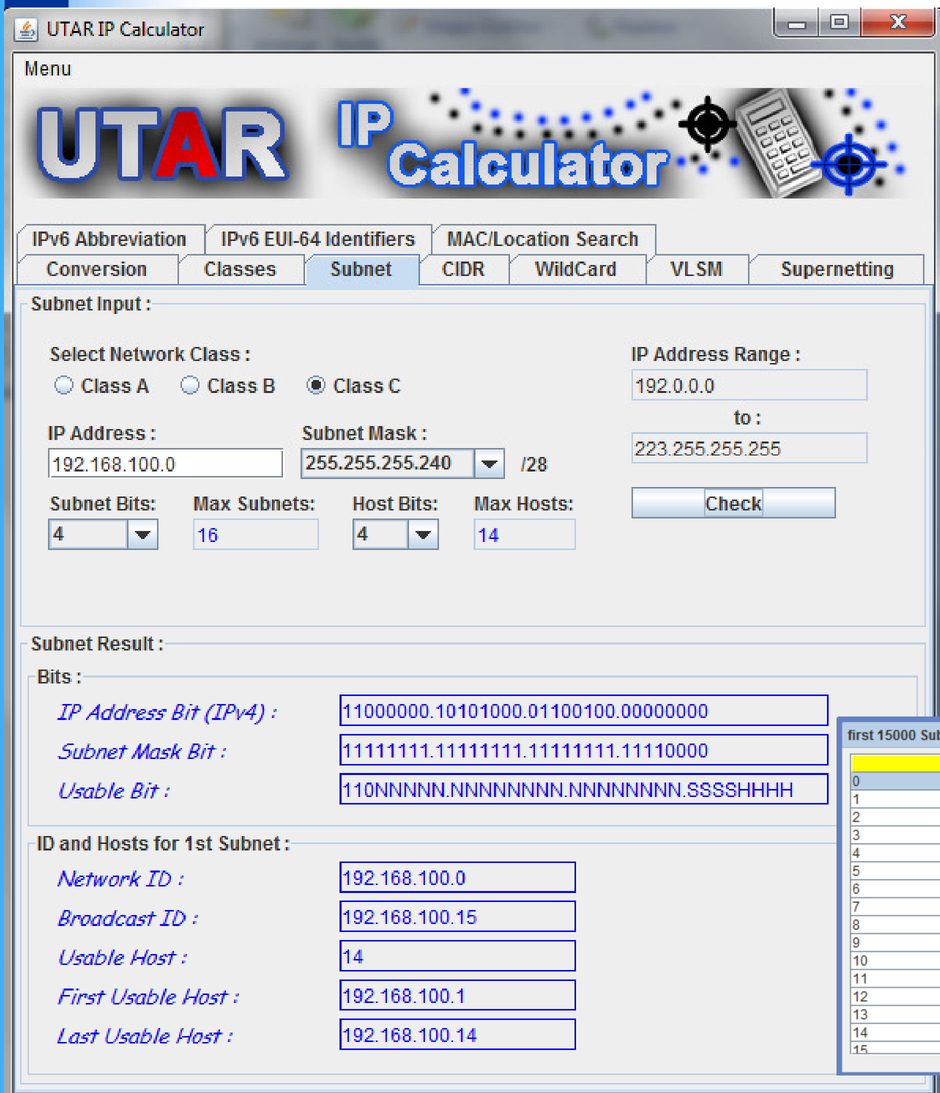
Using Bitcricket IP Calculator



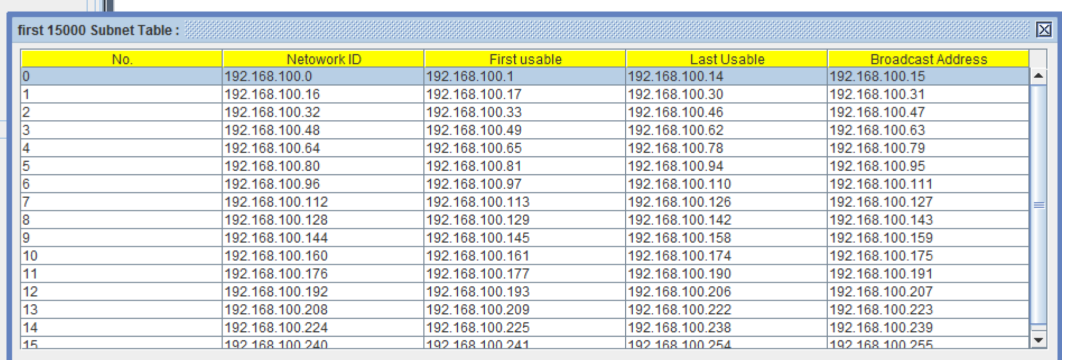
	Subnet ID	Host Addresses	Subnet Broadcast
0	192.168.100.0	192.168.100.1 - 192.168.100.14	192.168.100.15
1	192.168.100.16	192.168.100.17 - 192.168.100.30	192.168.100.31
2	192.168.100.32	192.168.100.33 - 192.168.100.46	192.168.100.47
3	192.168.100.48	192.168.100.49 - 192.168.100.62	192.168.100.63
4	192.168.100.64	192.168.100.65 - 192.168.100.78	192.168.100.79
5	192.168.100.80	192.168.100.81 - 192.168.100.94	192.168.100.95
6	192.168.100.96	192.168.100.97 - 192.168.100.110	192.168.100.111
7	192.168.100.112	192.168.100.113 - 192.168.100.126	192.168.100.127
8	192.168.100.128	192.168.100.129 - 192.168.100.142	192.168.100.143
9	192.168.100.144	192.168.100.145 - 192.168.100.158	192.168.100.159
10	192.168.100.160	192.168.100.161 - 192.168.100.174	192.168.100.175
11	192.168.100.176	192.168.100.177 - 192.168.100.190	192.168.100.191
12	192.168.100.192	192.168.100.193 - 192.168.100.206	192.168.100.207
13	192.168.100.208	192.168.100.209 - 192.168.100.222	192.168.100.223
14	192.168.100.224	192.168.100.225 - 192.168.100.238	192.168.100.239
15	192.168.100.240	192.168.100.241 - 192.168.100.254	192.168.100.255

- Key in the “Address”
  - In our case 192.168.100.0
- Choose the “Subnet Bits”
  - In our case: 4
- All other fields will be updated automatically.
  - Especially the subnet mask
- You can choose either one of the 4 field to design your subnet and the rest of the field will be updated:
  - Max Hosts
  - Host Bits
  - Max Subnets
  - Subnet bits

Using UTAR IP Calculator



- Click on “class C”
- Key in the “Address”
  - In our case 192.168.100.0
- Choose the “Subnet Bits”
  - In our case: 4
- Click on “Check”



first 15000 Subnet Table :

No.	Netowork ID	First usable	Last Usable	Broadcast Address
0	192.168.100.0	192.168.100.1	192.168.100.14	192.168.100.15
1	192.168.100.16	192.168.100.17	192.168.100.30	192.168.100.31
2	192.168.100.32	192.168.100.33	192.168.100.46	192.168.100.47
3	192.168.100.48	192.168.100.49	192.168.100.62	192.168.100.63
4	192.168.100.64	192.168.100.65	192.168.100.78	192.168.100.79
5	192.168.100.80	192.168.100.81	192.168.100.94	192.168.100.95
6	192.168.100.96	192.168.100.97	192.168.100.110	192.168.100.111
7	192.168.100.112	192.168.100.113	192.168.100.126	192.168.100.127
8	192.168.100.128	192.168.100.129	192.168.100.142	192.168.100.143
9	192.168.100.144	192.168.100.145	192.168.100.158	192.168.100.159
10	192.168.100.160	192.168.100.161	192.168.100.174	192.168.100.175
11	192.168.100.176	192.168.100.177	192.168.100.190	192.168.100.191
12	192.168.100.192	192.168.100.193	192.168.100.206	192.168.100.207
13	192.168.100.208	192.168.100.209	192.168.100.222	192.168.100.223
14	192.168.100.224	192.168.100.225	192.168.100.238	192.168.100.239
15	192.168.100.240	192.168.100.241	192.168.100.254	192.168.100.255

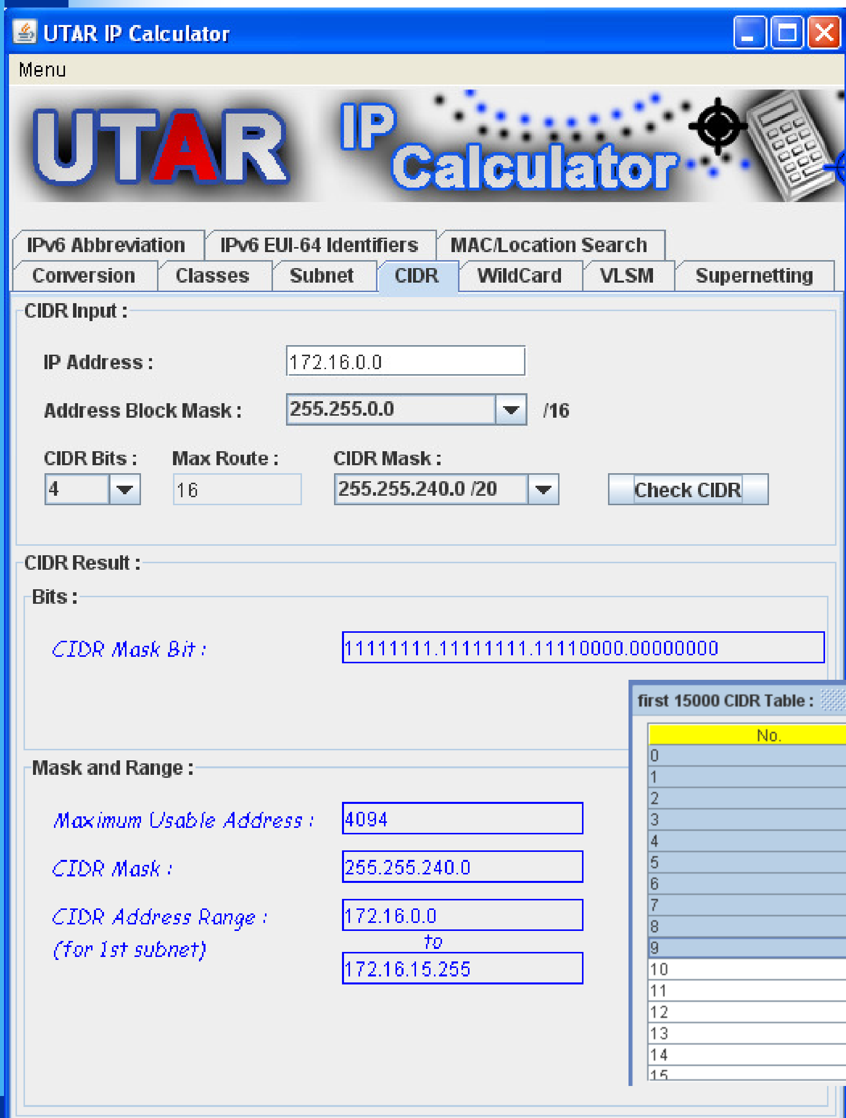
Quick Quiz: Subnetting

- You have address of 172.16.0.0 /16.
- You need nine subnets.
- What is the IP plan for these 9 subnets (assume NO ip subnet-zero)?

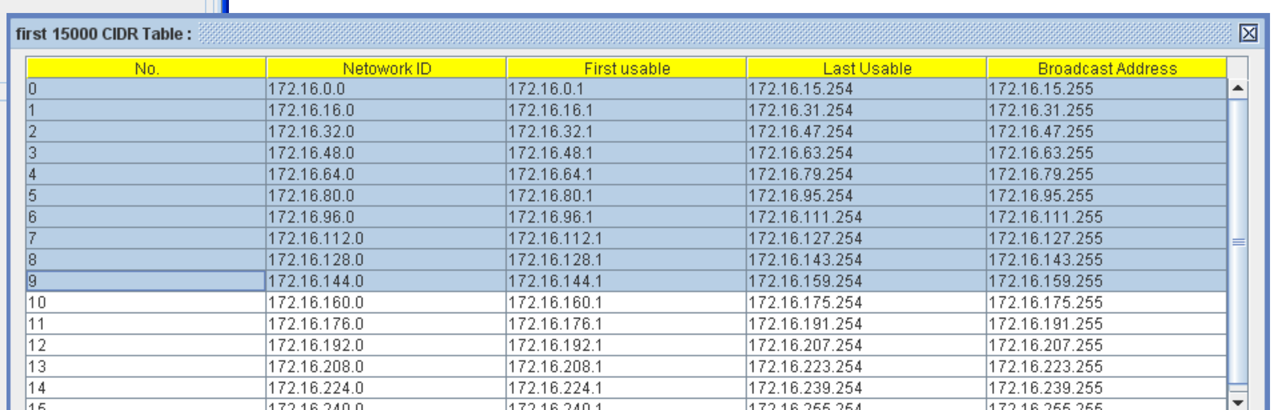
Answer

Subnet	Network Address (0000)	Range of Valid Hosts (0001–1110)	Broadcast Address (1111)
0 (0000) invalid	172.16.**0.0**	172.16.**0.1**–172.16.**15.254**	172.16.**15.255**
1 (0001)	172.16.**16.0**	172.16.**16.1**–172.16.**31.254**	172.16.**31.255**
2 (0010)	172.16.**32.0**	172.16.**32.1**–172.16.**47.254**	172.16.**47.255**
3 (0011)	172.16.**48.0**	172.16.**48.1**–172.16.**63.254**	172.16.**63.255**
4 (0100)	172.16.**64.0**	172.16.**64.1**–172.16.**79.254**	172.16.**79.255**
5 (0101)	172.16.**80.0**	172.16.**80.1**–172.16.**95.254**	172.16.**95.255**
6 (0110)	172.16.**96.0**	172.16.**96.1**–172.16.**111.254**	172.16.**111.255**
7 (0111)	172.16.**112.0**	172.16.**112.1**–172.16.**127.254**	172.16.**127.255**
8 (1000)	172.16.**128.0**	172.16.**128.1**–172.16.**143.254**	172.16.**143.255**
9 (1001)	172.16.**144.0**	172.16.**144.1**–172.16.**159.254**	172.16.**159.255**

Non-Classful Subnetting with Equal Subnet Mask using UTAR IP calc



- Use “CIDR”
- Key in “IP Address”
  - 172.16.0.0
- Key in “Address Block Mask”
  - /16
- Select “CIDR Bits”
  - 4 bits.
- Click in “Check CIDR”

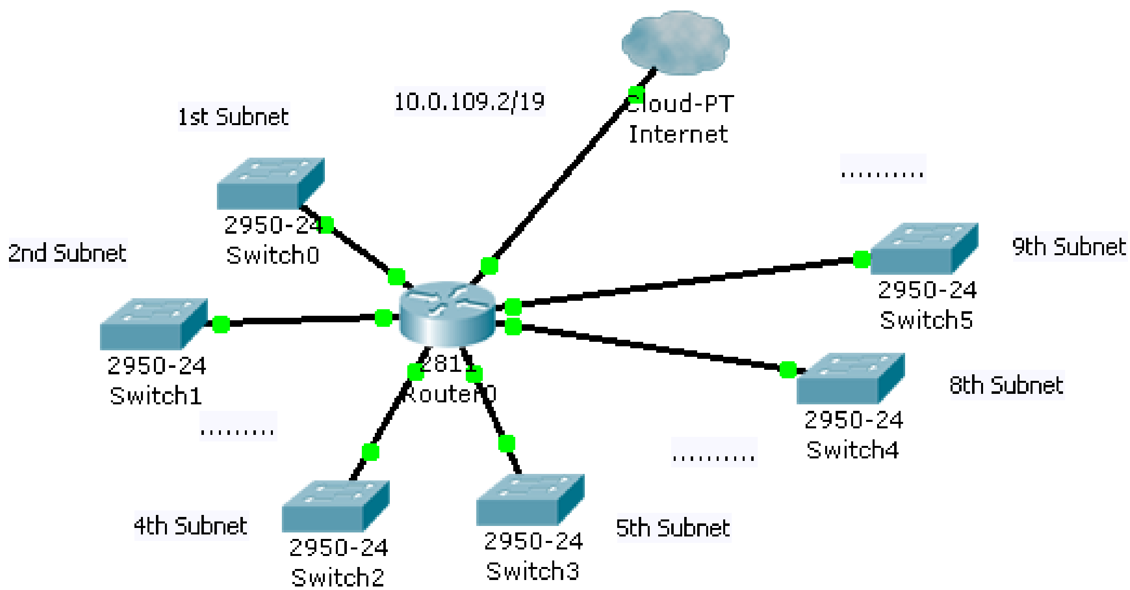


first 15000 CIDR Table :

No.	Netowork ID	First usable	Last Usable	Broadcast Address
0	172.16.0.0	172.16.0.1	172.16.15.254	172.16.15.255
1	172.16.16.0	172.16.16.1	172.16.31.254	172.16.31.255
2	172.16.32.0	172.16.32.1	172.16.47.254	172.16.47.255
3	172.16.48.0	172.16.48.1	172.16.63.254	172.16.63.255
4	172.16.64.0	172.16.64.1	172.16.79.254	172.16.79.255
5	172.16.80.0	172.16.80.1	172.16.95.254	172.16.95.255
6	172.16.96.0	172.16.96.1	172.16.111.254	172.16.111.255
7	172.16.112.0	172.16.112.1	172.16.127.254	172.16.127.255
8	172.16.128.0	172.16.128.1	172.16.143.254	172.16.143.255
9	172.16.144.0	172.16.144.1	172.16.159.254	172.16.159.255
10	172.16.160.0	172.16.160.1	172.16.175.254	172.16.175.255
11	172.16.176.0	172.16.176.1	172.16.191.254	172.16.191.255
12	172.16.192.0	172.16.192.1	172.16.207.254	172.16.207.255
13	172.16.208.0	172.16.208.1	172.16.223.254	172.16.223.255
14	172.16.224.0	172.16.224.1	172.16.239.254	172.16.239.255
15	172.16.240.0	172.16.240.1	172.16.255.254	172.16.255.255

Quick Quiz: Subnetting

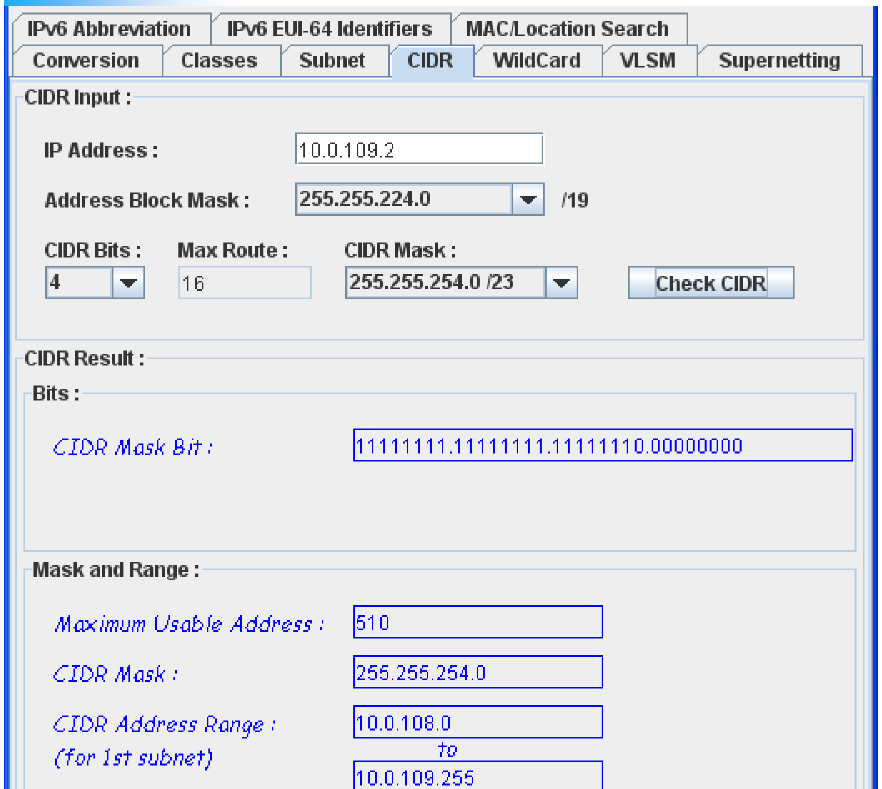
- You are being assigned to a network 10.0.109.2/19, and 4 bits are being allocated for subnets.
- You are asked to configure your PC to be the 8th host in the 5th subnet (0 subnet is not the 1st subnet).
- The gateway is the last usable host address.
- So, what will be your IP address, subnet mask, and gateway address?



Answer (using hand calculation)

- 10.0.109.0 = 10.0. 01101101. 2/19 =>
  10.0. 011 < 0110 >< 1.2>
- 10.0.011 < 0101>< 0.00001000> where
  <0101> = 5th subnet. <0.00001000> = 8th host.
- So IP = 10.0. "01101010 . 00001000" = 10.0.106.8
- Subnet mask = /23 = 255.255.254.0
- Gateway address =
  10.0. 011<0101><1 .11111110> = 10.0.107.254

Answer (using UTAR IP Calc)



- Key in 10.0.109.2, select /19, select 4 CIDR bits
- From IP calc, we know that 5th subnet is from 10.0.106.0 to 10.0.107.255
- 8th usable IP = <u>10.0.106.8</u>
  - 10.0.106.0 + 8
- Gateway (Last usable) = <u>10.0.107.254</u>
- New subnet mask: just copy from "CIDR Mask" = <u>/23</u>

No.	Netowork ID	First usable	Last Usable	Broadcast Address
0	10.0.96.0	10.0.96.1	10.0.97.254	10.0.97.255
1	10.0.98.0	10.0.98.1	10.0.99.254	10.0.99.255
2	10.0.100.0	10.0.100.1	10.0.101.254	10.0.101.255
3	10.0.102.0	10.0.102.1	10.0.103.254	10.0.103.255
4	10.0.104.0	10.0.104.1	10.0.105.254	10.0.105.255
5	10.0.106.0	10.0.106.1	10.0.107.254	10.0.107.255
6	10.0.108.0	10.0.108.1	10.0.109.254	10.0.109.255

VLSM

Variable Length Subnet Mask

CIDR & VLSM

- Prior to 1981, IP addresses used only the first 8 bits to specify the network portion of the address
- In 1981, RFC 791 modified the IPv4 32-bit address to allow for three different classes
- IP address space was depleting rapidly
  - the Internet Engineering Task Force (IETF) introduced Classless Inter-Domain Routing (CIDR)
  - CIDR uses Variable Length Subnet Masking (VLSM) to help conserve address space.
  - VLSM is simply subnetting a subnet

UCCN2243: VLSM (1)

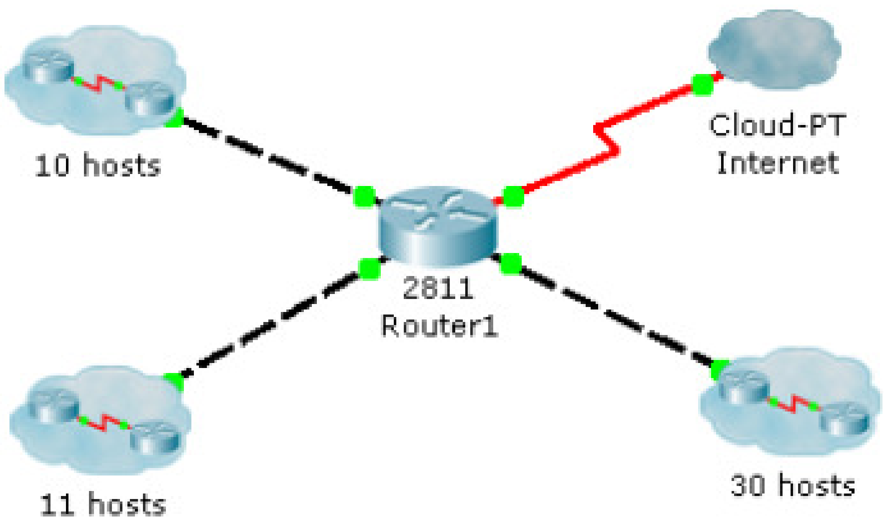
- Variable-length subnet masking (VLSM) is the more realistic way of subnetting a network to make for the most efficient use of all the bits.
- VLSM is the process of “subnetting a subnet” and using different subnet masks for different networks in your IP plan.
- What you have to remember is that you need to make sure that there is no overlap in any of the addresses.

UCCN2243: VLSM (2)

- Remember that when you perform classful (or what I sometimes call classical) subnetting, all subnets have the same number of hosts because they all use the same subnet mask.
- This leads to inefficiencies.
  - For example, if you borrow 4 bits on a Class C network, you end up with 14 valid subnets of 14 valid hosts. (instead of 16 subnets, if NO ip subnet-zero)
  - A serial link to another router only needs 2 hosts, but with classical subnetting, you end up wasting 12 of those hosts.

VLSM Example #1 (1)

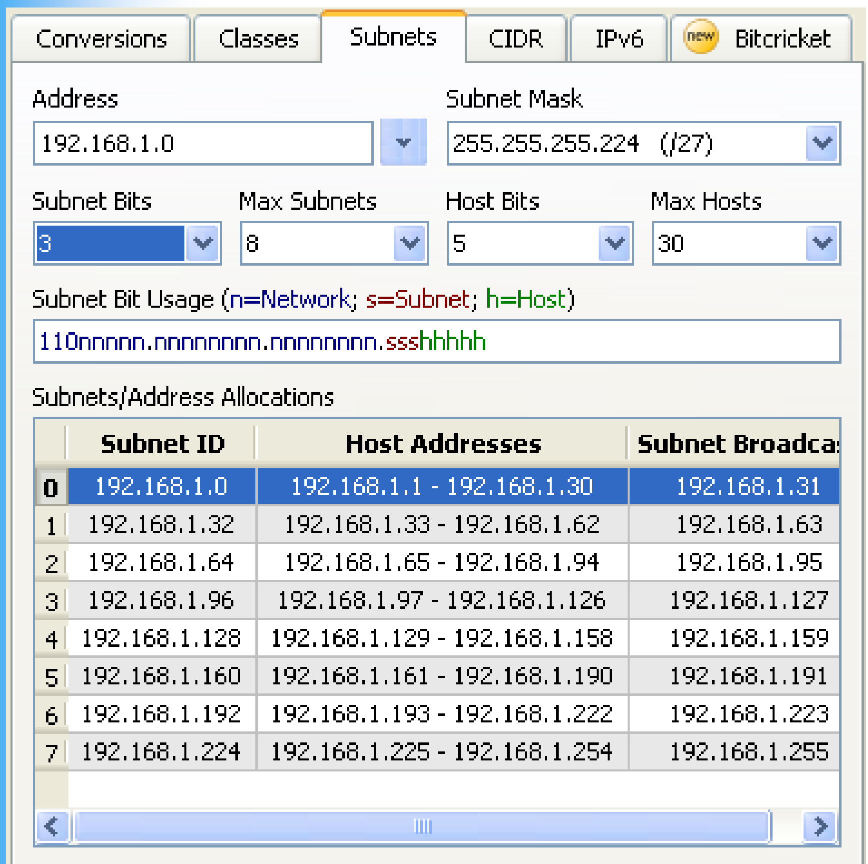
- Given the following network 192.168.1.0/24, how is it going to be subnetted?
  - We only need to take care of the 10 hosts subnet, 11 hosts subnet, and 30 hosts subnet.
  - Assume IP subnet-zero is implemented.
  - Assume that there won't be expansion on the number of host per subnet.



VLSM Example #1 (2)

- Using the traditional subnetting methodology.
  - There are 3 subnets and the maximum # host in a subnet is 30.
  - Either we have 2 subnet bits or 5 host bits.
  - If we choose 2 subnets bit design (6 bits host), there won't be much expansion on subnets but growth on hosts.
  - In this case, we choose 5 bits host design (3 subnet bits), as specified in the question.

VLSM Example #1 (3)


Conversions | Classes | Subnets | CIDR | IPv6 | Bitcricket

Address: 192.168.1.0

Subnet Mask: 255.255.255.224 (/27)

Subnet Bits: 3 | Max Subnets: 8 | Host Bits: 5 | Max Hosts: 30

Subnet Bit Usage (n=Network; s=Subnet; h=Host)

110nnnnn.nnnnnnnn.nnnnnnnn.ssshhhhh

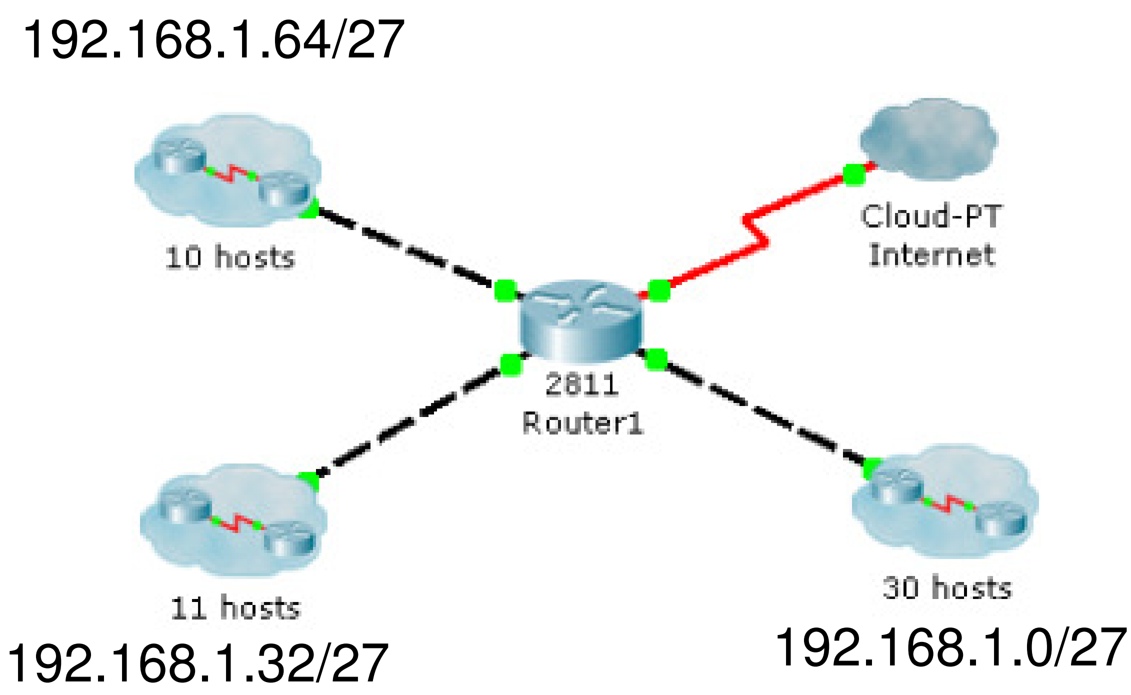
Subnets/Address Allocations

	Subnet ID	Host Addresses	Subnet Broadca:
0	192.168.1.0	192.168.1.1 - 192.168.1.30	192.168.1.31
1	192.168.1.32	192.168.1.33 - 192.168.1.62	192.168.1.63
2	192.168.1.64	192.168.1.65 - 192.168.1.94	192.168.1.95
3	192.168.1.96	192.168.1.97 - 192.168.1.126	192.168.1.127
4	192.168.1.128	192.168.1.129 - 192.168.1.158	192.168.1.159
5	192.168.1.160	192.168.1.161 - 192.168.1.190	192.168.1.191
6	192.168.1.192	192.168.1.193 - 192.168.1.222	192.168.1.223
7	192.168.1.224	192.168.1.225 - 192.168.1.254	192.168.1.255

- Using IP calculator to help our design.
- Since we have “ip subnet-zero, we can use the range
  - 192.168.1.1 to 192.168.1.30
  - 192.168.1.225 to 192.168.1.254
- You can pick any three of these 8 subnets.

VLSM Example #1 (4)

- With traditional classful subnetting



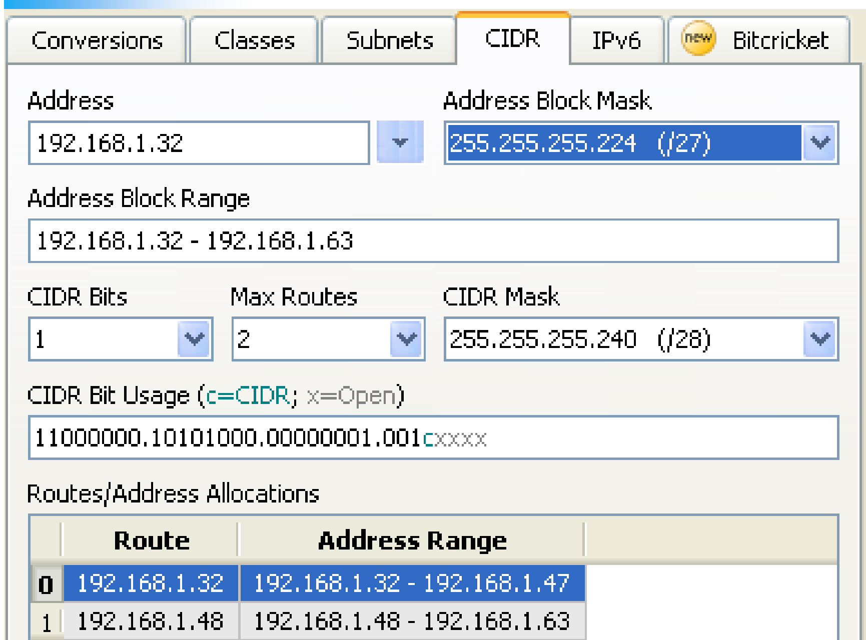
VLSM Example #1 (5)

- However, the traditional classful subnetting in this case, is not "IP address efficient".
- Given that there won't be expansion on the number of host per subnet.
- 10 hosts subnet and 11 hosts subnet will have around 20 IP addresses (per subnet) that are not utilized.

VLSM Example #1 (6)

- If we use VLSM scheme, this is how we are going to do it.
- 1st we choose the largest network 30 hosts
  - Host bits is still 5
  - We can still use the subnet of 192.168.1.1 to 192.168.1.30, which is 192.168.1.0/27
- 2nd we "group" the 10 hosts subnet and 11 hosts subnet into one "30 hosts" subnet.
  - We can use the subnet of 192.168.1.33 to 192.168.1.62, which is 192.168.1.32/27

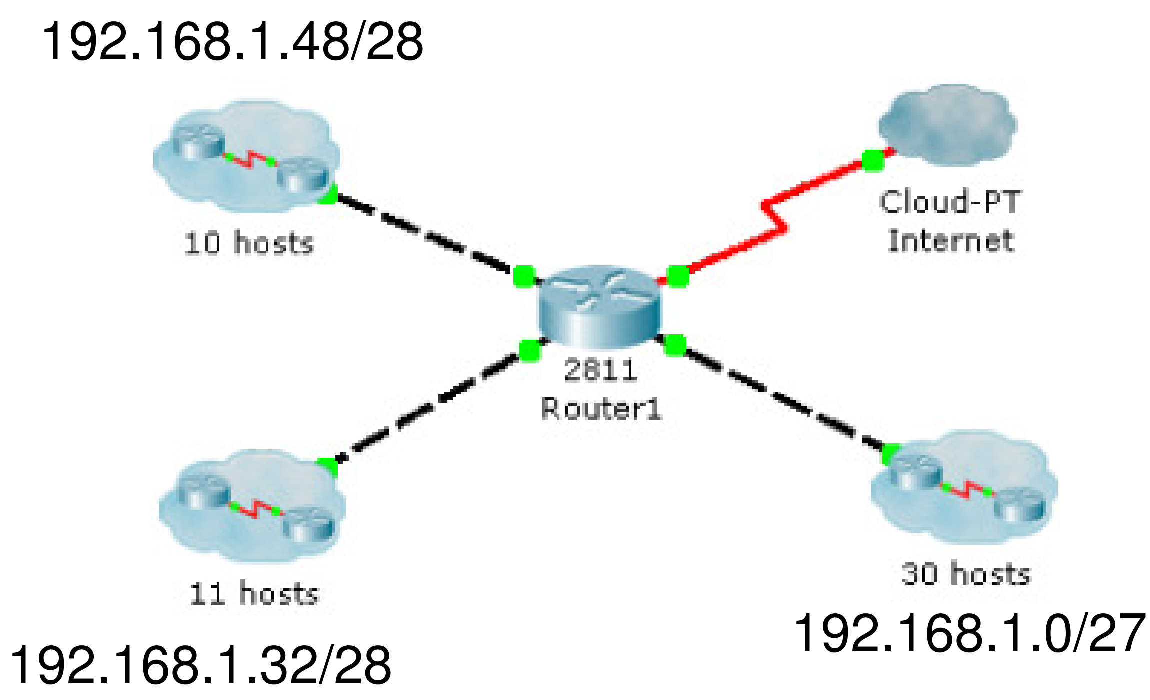
VLSM Example #1 (7)



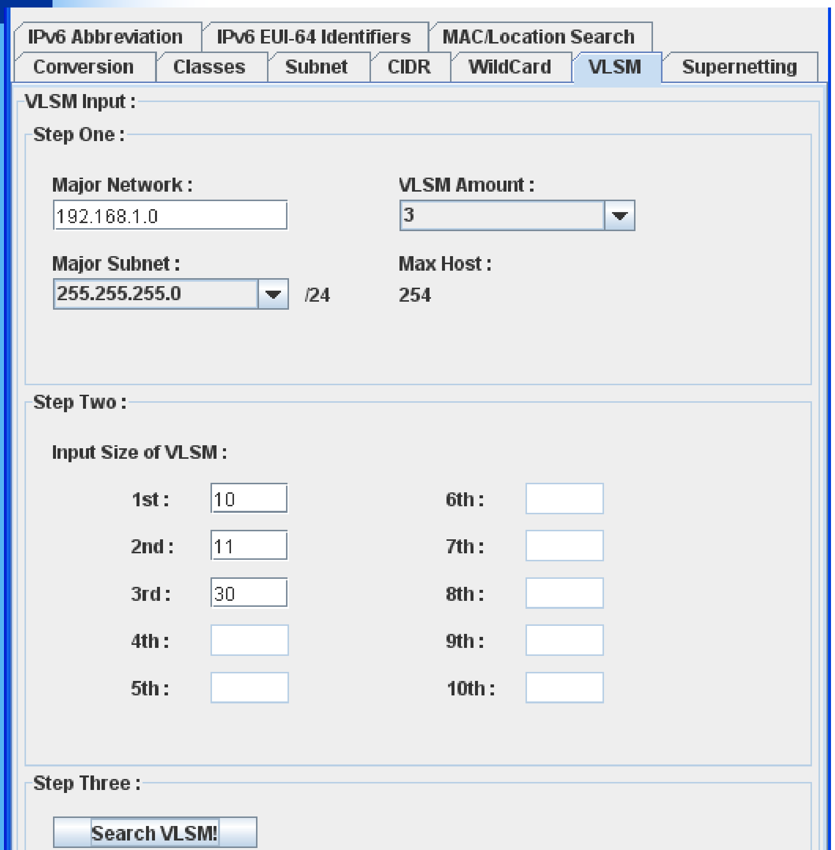
- Now, we perform subnetting the subnet.
- We subnet the 192.168.1.32/27 into two subnets.
- With the help of IP calculator (using CIDR), we have:
  - 192.168.1.32/28
    - Usable IP range 192.168.1.33 to 192.168.1.46
  - 192.168.1.48/28
    - Usable IP range 192.168.1.49 to 192.168.1.62

VLSM Example #1 (8)

- With VLSM subnetting



Using UTAR IP Calculator for VLSM

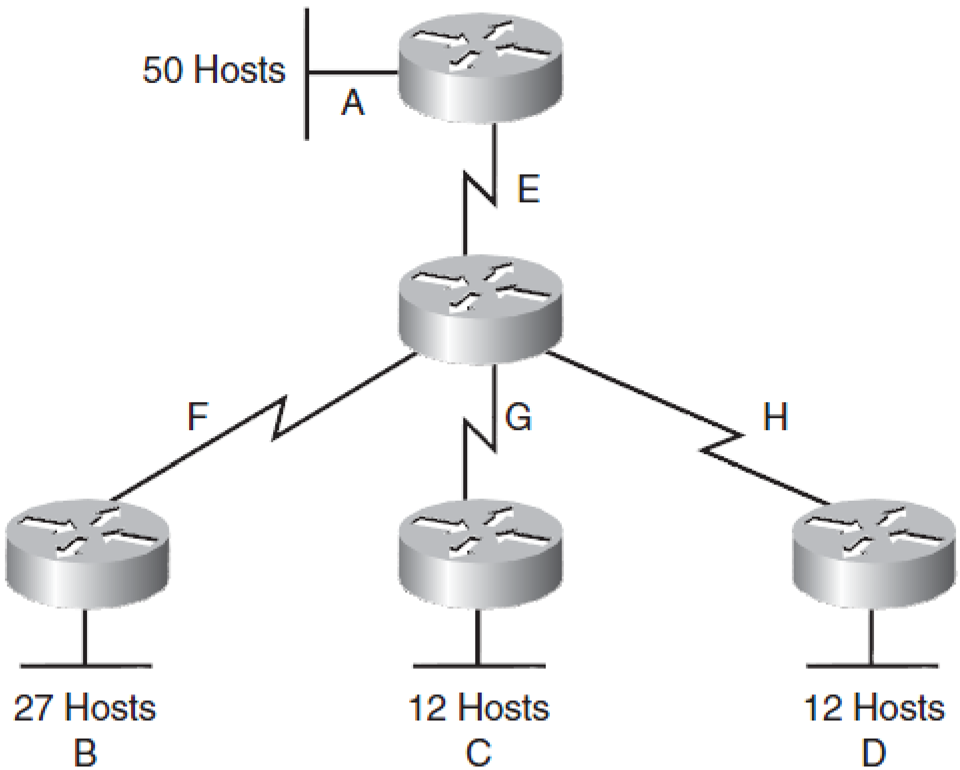


- Click on "VLSM" tab.
- Key in "192.168.1.0"
- Select /24
- Select 3 subnets
  - VLSM Amount
- Key in the host numbers in 3 subnets.
  - 10,11,30
- Click on "Search VLSM"

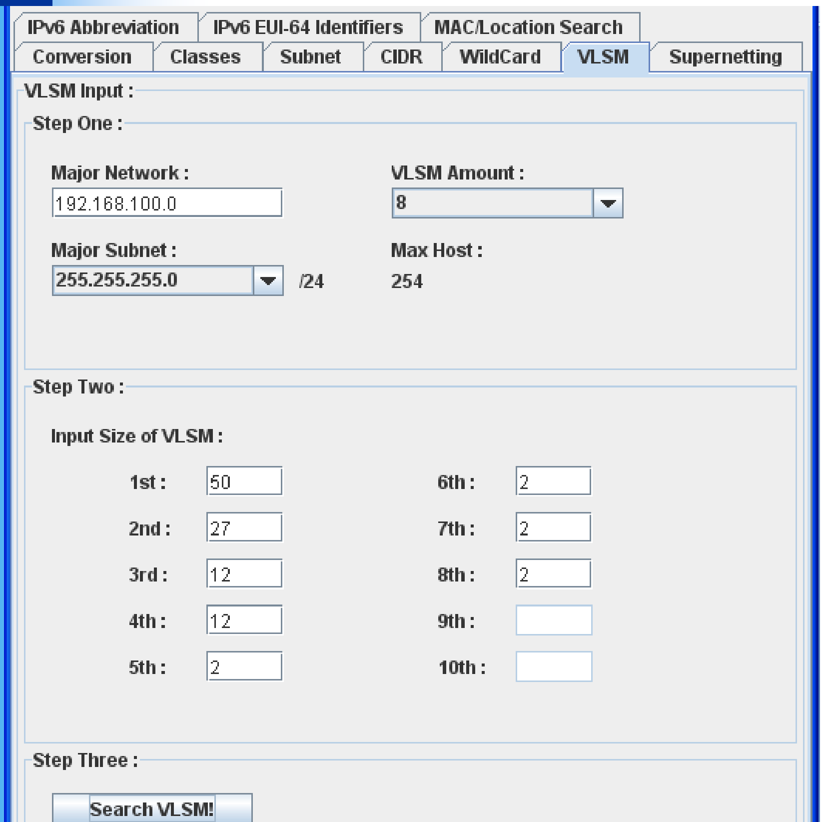
Subnet No.	Size	Allocated Size	Network Add	Mask	Dec Mask	Range	Broadcast Add
1	30	30	192.168.1.0	/27	255.255.255.224	192.168.1.1 to 192.168.1.30	192.168.1.31
2	11	14	192.168.1.32	/28	255.255.255.240	192.168.1.33 to 192.168.1.46	192.168.1.47
3	10	14	192.168.1.48	/28	255.255.255.240	192.168.1.49 to 192.168.1.62	192.168.1.63

UCCN2243: VLSM Example (1)

- A Class C network—192.168.100.0/24—is assigned. You need to create an IP plan for this network using VLSM.



The "Lazy" way – UTAR IP Calc



- Key in "192.168.100.0" and /24
- You need 8 subnets
  - A,B,C,D,E,F,G,H
- Key in the host numbers in the subnets
  - 50,27,12,12,2,2,2,2
  - 2 => two IP in point-to-point network.
- Click on "Search VLSM"

Subnet No.	Size	Allocated Size	Network Add	Mask	Dec Mask	Range	Broadcast Add
1	50	62	192.168.100.0	/26	255.255.255.192	192.168.100.1 to 192.168.100...	192.168.100.63
2	27	30	192.168.100.64	/27	255.255.255.224	192.168.100.65 to 192.168.10...	192.168.100.95
3	12	14	192.168.100.96	/28	255.255.255.240	192.168.100.97 to 192.168.10...	192.168.100.111
4	12	14	192.168.100.112	/28	255.255.255.240	192.168.100.113 to 192.168.1...	192.168.100.127
5	2	2	192.168.100.128	/30	255.255.255.252	192.168.100.129 to 192.168.1...	192.168.100.131
6	2	2	192.168.100.132	/30	255.255.255.252	192.168.100.133 to 192.168.1...	192.168.100.135
7	2	2	192.168.100.136	/30	255.255.255.252	192.168.100.137 to 192.168.1...	192.168.100.139
8	2	2	192.168.100.140	/30	255.255.255.252	192.168.100.141 to 192.168.1...	192.168.100.143

UCCN2243: VLSM Example (2)

- The steps to create an IP plan using VLSM for the network illustrated are as follows:
  - Step 1: Determine how many H bits will be needed to satisfy the largest network.
  - Step 2: Pick a subnet for the largest network to use.
  - Step 3: Pick the next largest network to work with.
  - Step 4: Pick the third largest network to work with.
  - Step 5: Determine network numbers for serial links.

UCCN2243: VLSM Example (3)

Step 1 Determine How Many H Bits Will Be Needed to Satisfy the *Largest* Network

A is the largest network with 50 hosts. Therefore, you need to know how many H bits will be needed:

If $2^H - 2$ = Number of valid hosts per subnet
Then $2^H - 2 \geq 50$
Therefore $H = 6$ (6 is the smallest valid value for H)

You need 6 H bits to satisfy the requirements of Network A.

If you need 6 H bits and you started with 8 N bits, you are left with $8 - 6 = 2$ N bits to create subnets:

Started with: NNNNNNNN (these are the 8 bits in the fourth octet)
Now have: NNHHHHHH

All subnetting will now have to start at this reference point, to satisfy the requirements of Network A.

UCCN2243: VLSM Example (4)

Step 2 Pick a Subnet for the Largest Network to Use

You have 2 N bits to work with, leaving you with 2^N or 2^2 or 4 subnets to work with:

NN = 00HHHHHH (The Hs = The 6 H bits you need for Network A)
01HHHHHH
10HHHHHH
11HHHHHH

If you add all zeros to the H bits, you are left with the network numbers for the four subnets:

00000000 = .0
01000000 = .64
10000000 = .128
11000000 = .192

All of these subnets will have the same subnet mask, just like in classful subnetting.

Two borrowed H bits means a subnet mask of

11111111.11111111.11111111.11000000

or

255.255.255.192

or

/26

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Next page

UCCN2243: VLSM Example (5)

The /x notation represents how to show different subnet masks when using VLSM.

/8 means that the first 8 bits of the address are network; the remaining 24 bits are H bits.

/24 means that the first 24 bits are network; the last 8 are host. This is either a traditional default Class C address, or a traditional Class A network that has borrowed 16 bits, or even a traditional Class B network that has borrowed 8 bits!

Pick *one* of these subnets to use for Network A. The rest of the networks will have to use the other three subnets.

For purposes of this example, pick the .64 network.

00000000 =	.0	
01000000 =	.64	Network A
10000000 =	.128	
11000000 =	.192	

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UCCN2243: VLSM Example (6)

Step 3 Pick the Next Largest Network to Work With

Network B = 27 hosts

Determine the number of H bits needed for this network:

$$2^H - 2 \geq 27$$
$$H = 5$$

You need 5 H bits to satisfy the requirements of Network B.

You started with a pattern of 2 N bits and 6 H bits for Network A. You have to maintain that pattern.

Pick one of the remaining /26 networks to work with Network B.

For the purposes of this example, select the .128/26 network:

1**0000000**

But you need only 5 H bits, not 6. Therefore, you are left with

10**N00000**

where

10 represents the original pattern of subnetting.
N represents the extra bit.
00000 represents the 5 H bits you need for Network B.

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Next page

UCCN2243: VLSM Example (7)

Because you have this extra bit, you can create two smaller subnets from the original subnet:

10000000
10100000

Converted to decimal, these subnets are as follows:

10000000 =.128
10100000 =.160

You have now subnetted a subnet! This is the basis of VLSM.

Each of these sub-subnets will have a new subnet mask. The original subnet mask of /24 was changed into /26 for Network A. You then take one of these /26 networks and break it into two /27 networks:

10000000 and 10100000 both have 3 N bits and 5 H bits.

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Next page

UCCN2243: VLSM Example (8)

The mask now equals:

11111111.11111111.11111111.11100000

or

255.255.255.224

or

/27

Pick one of these new sub-subnets for Network B:

10000000 /27 = Network B

Use the remaining sub-subnet for future growth, or you can break it down further if needed.

You want to make sure the addresses are not overlapping with each other. So go back to the original table.

00000000 =	.0/26	
01000000 =	.64/26	Network A
10000000 =	.128/26	
11000000 =	.192/26	

Continue
Next page

UCCN2243: VLSM Example (9)

You can now break the .128/26 network into two smaller /27 networks and assign Network B.

00000000 =	.0/26	
01000000 =	.64/26	Network A
10000000 =	.128/26	Cannot use because it has been subnetted
10000000 =	.128/27	Network B
10100000 =	.160/27	
11000000 =	.192/26	

The remaining networks are still available to be assigned to networks or subnetted further for better efficiency.

UCCN2243: VLSM Example (10)

Step 4 Pick the Third Largest Network to Work With

Networks C and Network D = 12 hosts each

Determine the number of H bits needed for these networks:

$$2^H - 2 \geq 12$$
$$H = 4$$

You need 4 H bits to satisfy the requirements of Network C and Network D.

You started with a pattern of 2 N bits and 6 H bits for Network A. You have to maintain that pattern.

You now have a choice as to where to put these networks. You could go to a different /26 network, or you could go to a /27 network and try to fit them into there.

For the purposes of this example, select the other /27 network—.160/27:

101**00000** (The 1 in the third bit place is no longer bold, because it is part of the N bits.)

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Next page

UCCN2243: VLSM Example (11)

But you only need 4 H bits, not 5. Therefore, you are left with

101**N0000**

where

10 represents the original pattern of subnetting.
N represents the extra bit you have.
00000 represents the 5 H bits you need for Network B.

Because you have this extra bit, you can create two smaller subnets from the original subnet:

101**00000**
101**10000**

Converted to decimal, these subnets are as follows:

101**00000** = .160
101**10000** = .176

These new sub-subnets will now have new subnet masks. Each sub-subnet now has 4 N bits and 4 H bits, so their new masks will be

11111111.11111111.11111111.11110000

or

255.255.255.240

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Next page

UCCN2243: VLSM Example (12)

00000000 =	.0/26	
01000000 =	.64/26	Network A
10000000 =	.128/26	Cannot use because it has been subnetted
10000000 =	.128/27	Network B
10100000 =	.160/27	Cannot use because it has been subnetted
10100000	.160/28	Network C
10110000	.176/28	Network D
11000000 =	.192/26	

You have now used two of the original four subnets to satisfy the requirements of four networks. Now all you need to do is determine the network numbers for the serial links between the routers.

UCCN2243: VLSM Example (13)

Step 5 Determine Network Numbers for Serial Links

All serial links between routers have the same property in that they only need two addresses in a network—one for each router interface.

Determine the number of H bits needed for these networks:

$$2^H - 2 \geq 2$$
$$H = 2$$

You need 2 H bits to satisfy the requirements of Networks E, F, G, and H.

You have two of the original subnets left to work with.

For the purposes of this example, select the .0/26 network:

00**000000**

But you need only 2 H bits, not 6. Therefore, you are left with

00**NNNN00**

where

00 represents the original pattern of subnetting.
NNNN represents the extra bits you have.
00 represents the 2 H bits you need for the serial links.

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UCCN2243: VLSM Example (14)

Because you have 4 **N** bits, you can create 16 sub-subnets from the original subnet:

00000000 = .0/30
00000100 = .4/30
00001000 = .8/30
00001100 = .12/30
00010000 = .16/30
.
.
.
00111000 = .56/30
00111100 = .60/30

Continue
Next page

You need only four of them. You can hold the rest for future expansion or recombine them for a new, larger subnet:

00010000 = .16/30
.
.
.
00111000 = .56/30
00111100 = .60/30

All these can be recombined into the following:

00010000 = .16/28

UTAR
UNIVERSITI TUNKU ABDUL RAHMAN

UCCN2243: VLSM Example (15)

Going back to the original table, you now have the following:

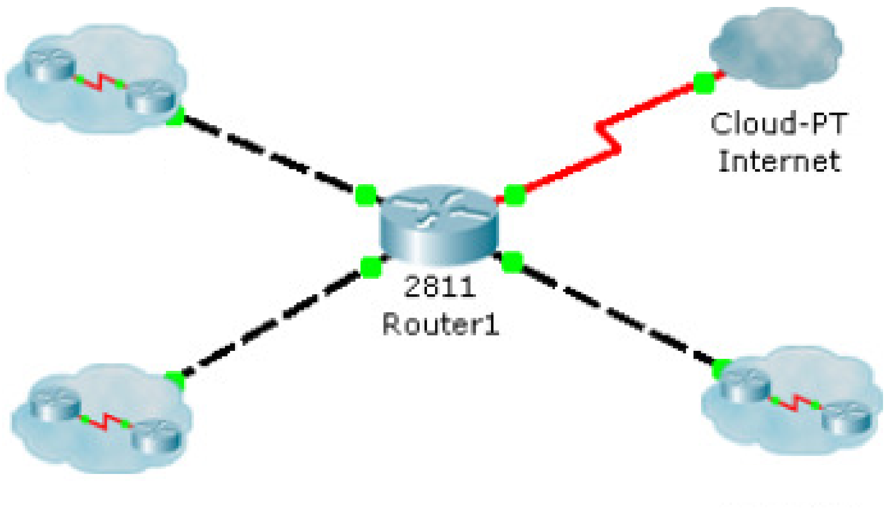
00000000 =	.0/26	Cannot use because it has been subnetted
00000000 =	.0/30	Network E
00000100 =	.4/30	Network F
00001000 =	.8/30	Network G
00001100 =	.12/30	Network H
00010000 =	.16/28	Future growth
01000000 =	.64/26	Network A
10000000 =	.128/26	Cannot use because it has been subnetted
10000000 =	.128/27	Network B
10100000 =	160/27	Cannot use because it has been subnetted
10100000	160/28	Network C
10110000	176/28	Network D
11000000 =	.192/26	Future growth

UCCN2243: VLSM Conclusion

- Looking at the plan, you can see that no number is used twice.
- You have now created an IP plan for the network and have made the plan as efficient as possible, wasting no addresses in the serial links and leaving room for future growth.
- This is the power of VLSM.

Quick Quiz: Subnetting

- Given the following network 192.168.1.0/24, how is it going to be subnetted with 50 hosts, 60 hosts, and 70 hosts?



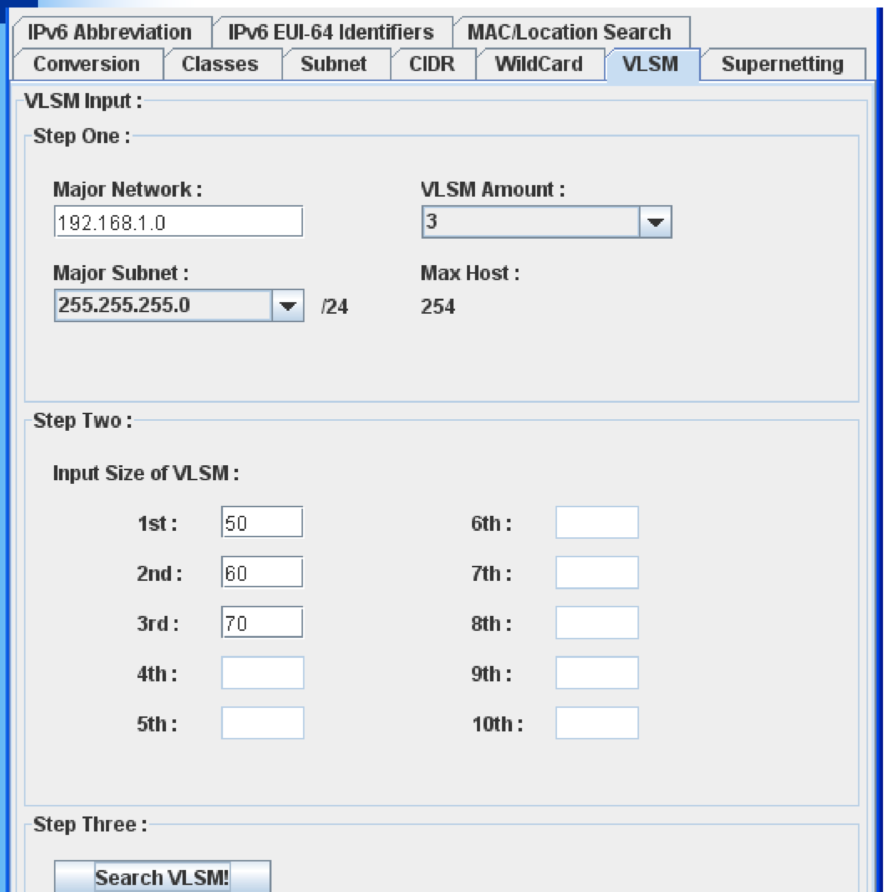
Answer 1: Traditional subnetting

- With traditional method, the network requires 2 subnet bits (assume with ip subnet-zero).
- That leave only 6 host bits (max # of host 62), then we can't "fit" 70 hosts with only 6 host bits.
- Therefore, we can't subnet this network with traditional method.

Answer 2: VLSM subnetting

- Start with largest network, 70 hosts, require 7 host bits, left with 1 subnet bit.
  - Which make the new subnet mask /25
  - We can put the 70 hosts in either 192.168.1.<0xxxxxxx>/25 (192.168.1.0/25) or 192.168.1.<1xxxxxxx>/25 (192.168.1.128/25)
    - We choose the first subnet, which is 192.168.1.0/25
    - The rest of the two subnets will be fit into 192.168.1.128/25
- Follow with the second largest network, 60 hosts, require 6 host bits. We need to perform subnetting the subnet (192.168.1.128/25), which leave us with 1 new subnet bit
  - Which make the new subnet mask /26
  - We can put the 60 hosts in either 192.168.1.<1'0'xxxxxx>/26 (192.168.1.128/26) or 192.168.1.<1'1'xxxxxx>/26 (192.168.1.192/26)
    - We choose the 60 hosts in the subnet of 192.168.1.128/26
    - We put the remaining 50 hosts in the subnet of 192.168.1.192/26

Using UTAR IP Calc

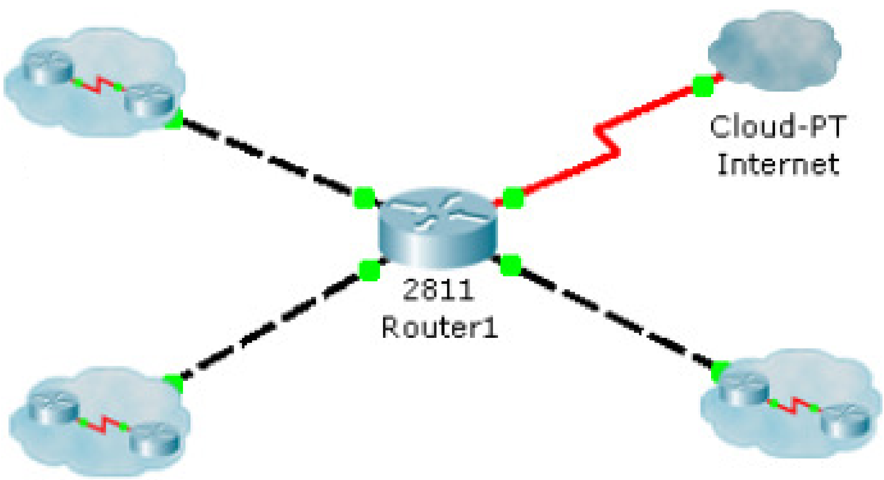


- You should learn how to key in this.

Subnet No.	Size	Allocated Size	Network Add	Mask	Dec Mask	Range	Broadcast Add
1	70	126	192.168.1.0	/25	255.255.255.128	192.168.1.1 to 192.168.1.126	192.168.1.127
2	60	62	192.168.1.128	/26	255.255.255.192	192.168.1.129 to 192.168.1.190	192.168.1.191
3	50	62	192.168.1.192	/26	255.255.255.192	192.168.1.193 to 192.168.1.254	192.168.1.255

Quick Quiz: Subnetting (again)

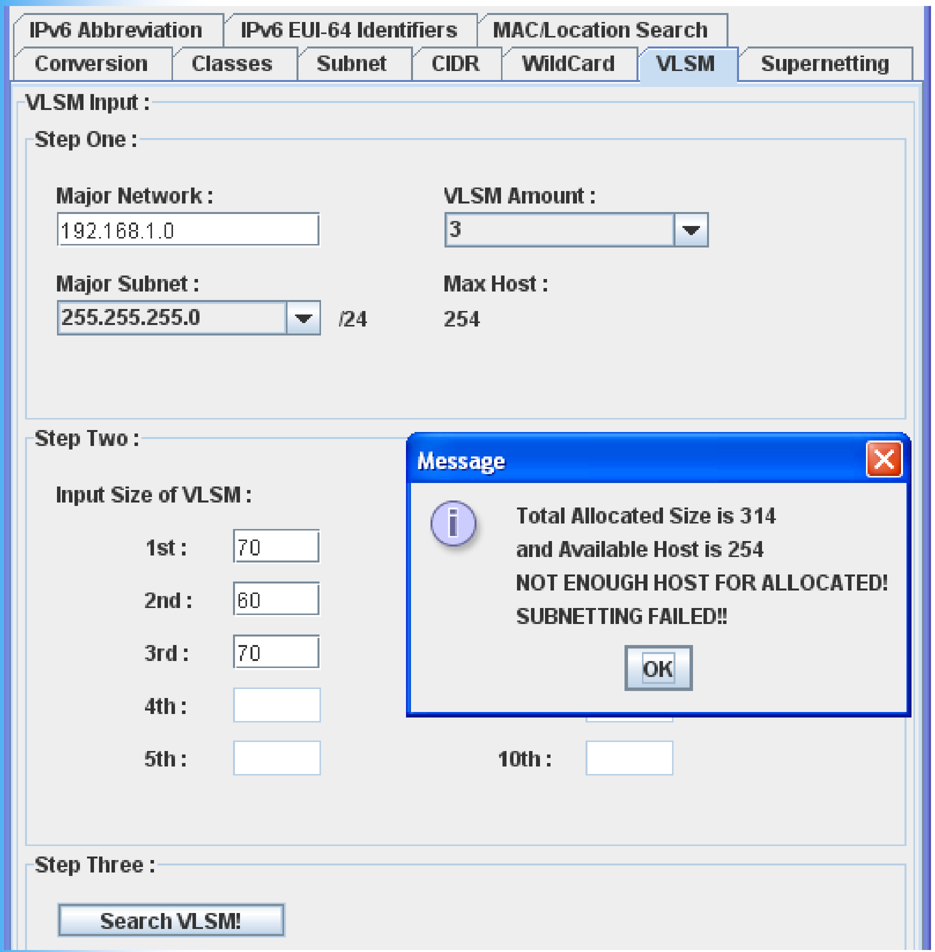
- Given the following network 192.168.1.0/24, how is it going to be subnetted with 70 hosts, 60 hosts, and 70 hosts? (previously it was 50, 60 and 70 hosts)



Answer:

- This network can NOT be subnetted with traditional subnetting or VLSM with 192.168.1.0/24.
  - Though 70+60+70 = 200 hosts (which are less than 254 host, which is 8 host bits)
- Reason:
  - There are two largest subnet with 70 hosts which require 7 host bits, which leave us with 1 bit subnet (# of subnets = 2).
  - After fitting the two 70 hosts subnet, there is no more subnet left for the 60 hosts.
  - You can either fit in 2 of the 3 subnets.

Using UTAR IP Calc



- In this special case, even UTAR IP calculator can't perform the VLSM subnetting.

UCCN2243: Pros and Cons of subnetting

- Pros:
  - Better security and management
  - More host IP being utilized within the subnet.
- Cons:
  - Waste some host IP in the form of network address, broadcast address, and gateway IPs
  - Network design is more complex in order to perform the IP planning.

Supernetting, CIDR & Route Summarization

They are using the same computation

UCCN2243: Route Summarization

- Route summarization, is needed to reduce the number of routes that a router advertises to its neighbor.
  - Router summarization make the “combined” subnets appear to be a “supernet”
- For every route you advertise, the size of the update and routing table grows.
- Route summarization greatly reduces the size of the routing table
  - It has been said that if there were no route summarization, the Internet backbone would have collapsed from the sheer size of its own routing tables back in 1997.

CIDR

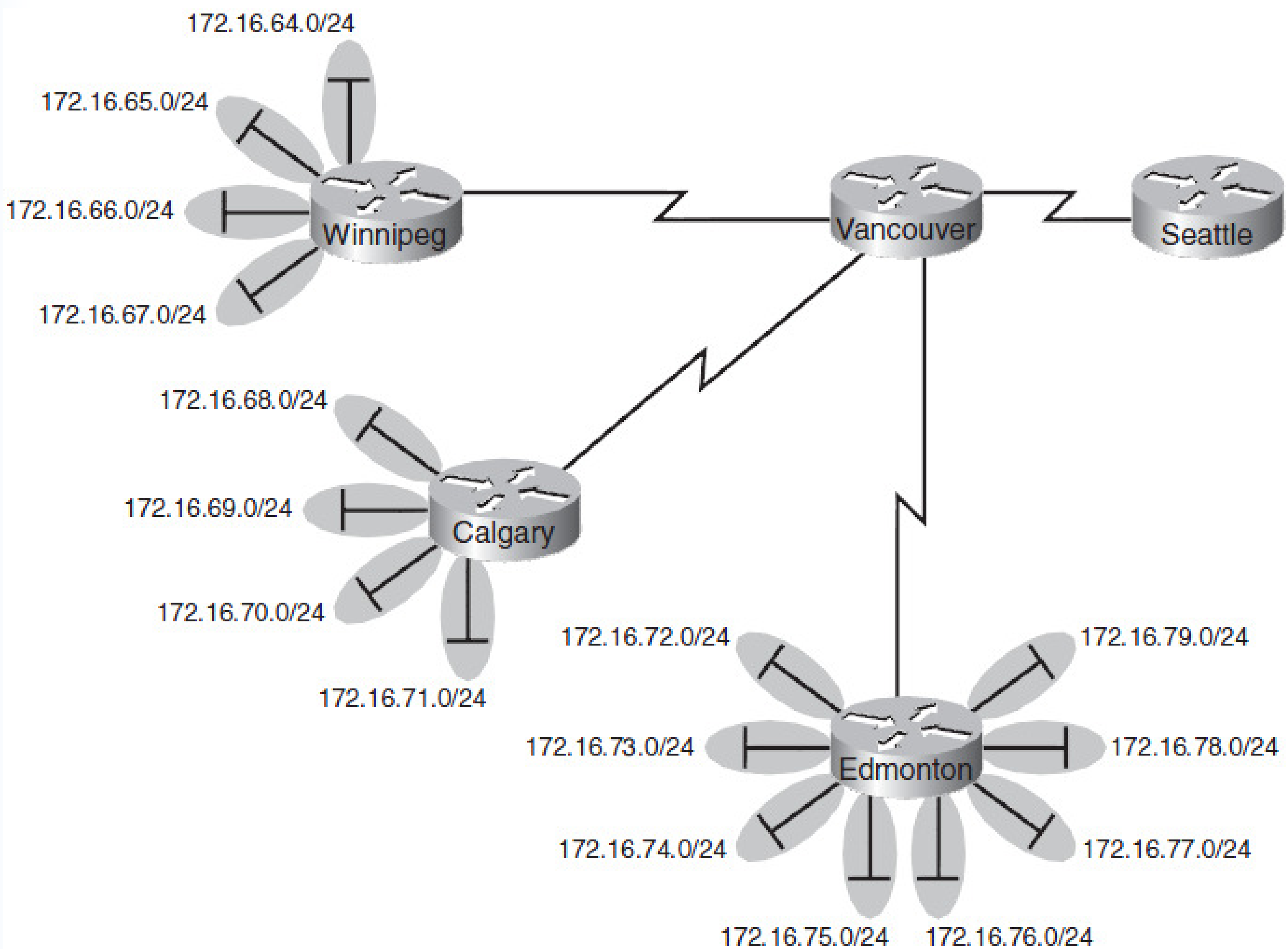
- CIDR (Classless Inter Domain Routing) is also known as supernetting as it effectively allows multiple subnets to be grouped together for network routing.

- CIDR was adopted to help ease the load imposed on internet and large network backbone routers by the increasing size of routing tables.
  - Large routing tables have several adverse effects:
  - Routers require more memory in order to store and manipulate their routing tables which increases operation costs.
  - Routing latency is increased due to the large amount of data contained in the routing tables.
  - Network bandwidth usage is increased by routing updates when routers exchange their routing tables.

CIDR

- CIDR (Classless Inter-Domain Routing) encompasses:
  – the VLSM technique of specifying arbitrary-length prefixes.
    - An address in CIDR notation is written with a suffix indicating the number of bits in the prefix, such as 192.168.0.0/16, where /16 is the suffix, and 192.168.0.0 is the prefix.
  – the aggregation of multiple contiguous prefixes into supernets, thus reducing the number of entries in the global routing table.
    - Aggregation hides multiple levels of subnetting from the Internet routing table, and reverses the process of subnetting with VLSM.

Route Summarization Example (1)



Route Summarization Example (2)

Step 1: Summarize Winnipeg's Routes

To do this, you need to look at the routes in binary to see if there are any specific bit patterns that you can use to your advantage. What you are looking for are common bits on the network side of the addresses. Because all of these networks are /24 networks, you want to see which of the first 24 bits are common to all four networks.

172.16.64.0 = *10101100.00010000.010000*00.00000000
172.16.65.0 = *10101100.00010000.010000*01.00000000
172.16.66.0 = *10101100.00010000.010000*10.00000000
172.16.67.0 = *10101100.00010000.010000*11.00000000
Common bits: *10101100.00010000.010000*xx

You see that the first 22 bits of the four networks are common. Therefore, you can summarize the four routes by using a subnet mask that reflects that the first 22 bits are common. This is a /22 mask, or 255.255.252.0. You are left with the summarized address of

172.16.64.0/22

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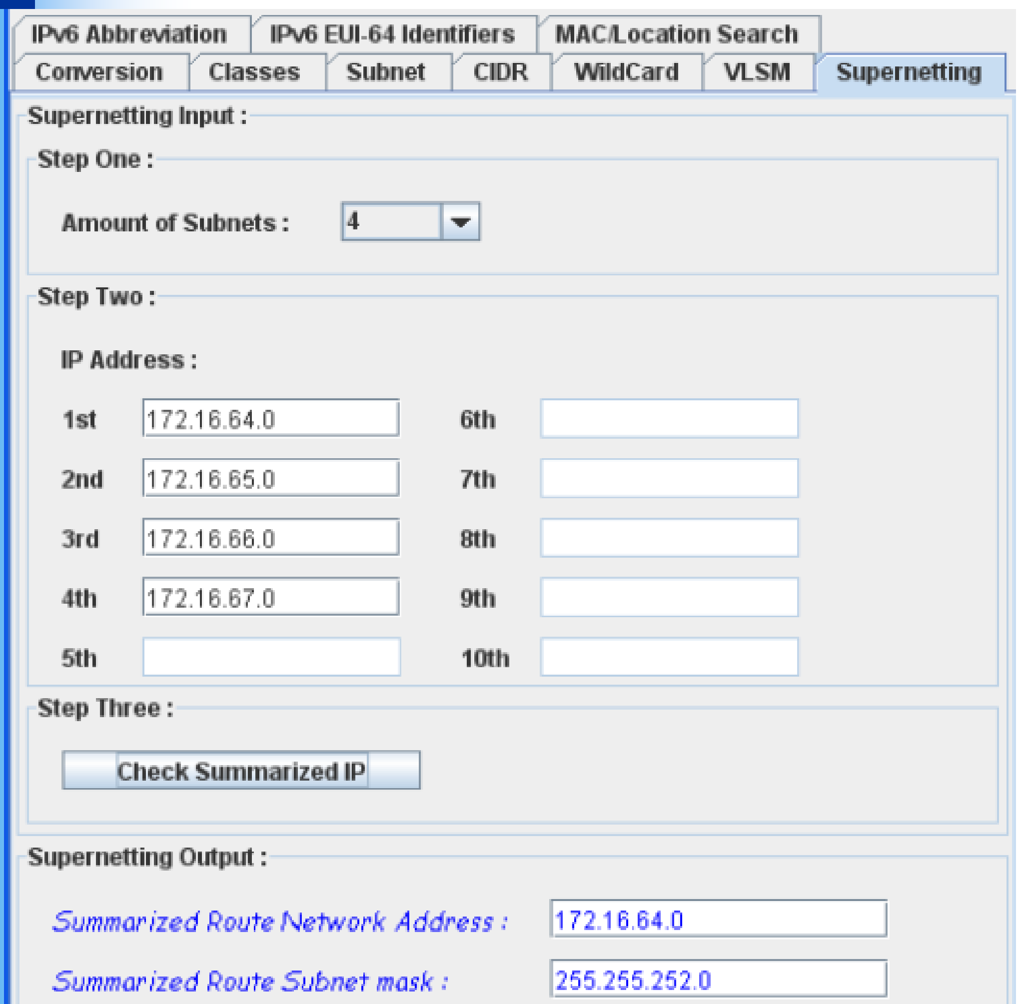
Using UTAR IP Calc



- Click on "Supernetting" tab
- Select "Amount of Subnets = 4
- Key in all subnet ID.
  - 172.16.64.0
  - 172.16.65.0
  - 172.16.66.0
  - 172.16.67.0
- Click on "Check Summarized IP".
- Answer:
  - 172.16.64.0
  - 255.255.252.0 => /22

Route Summarization Example (3)

Step 2: Summarize Calgary's Routes

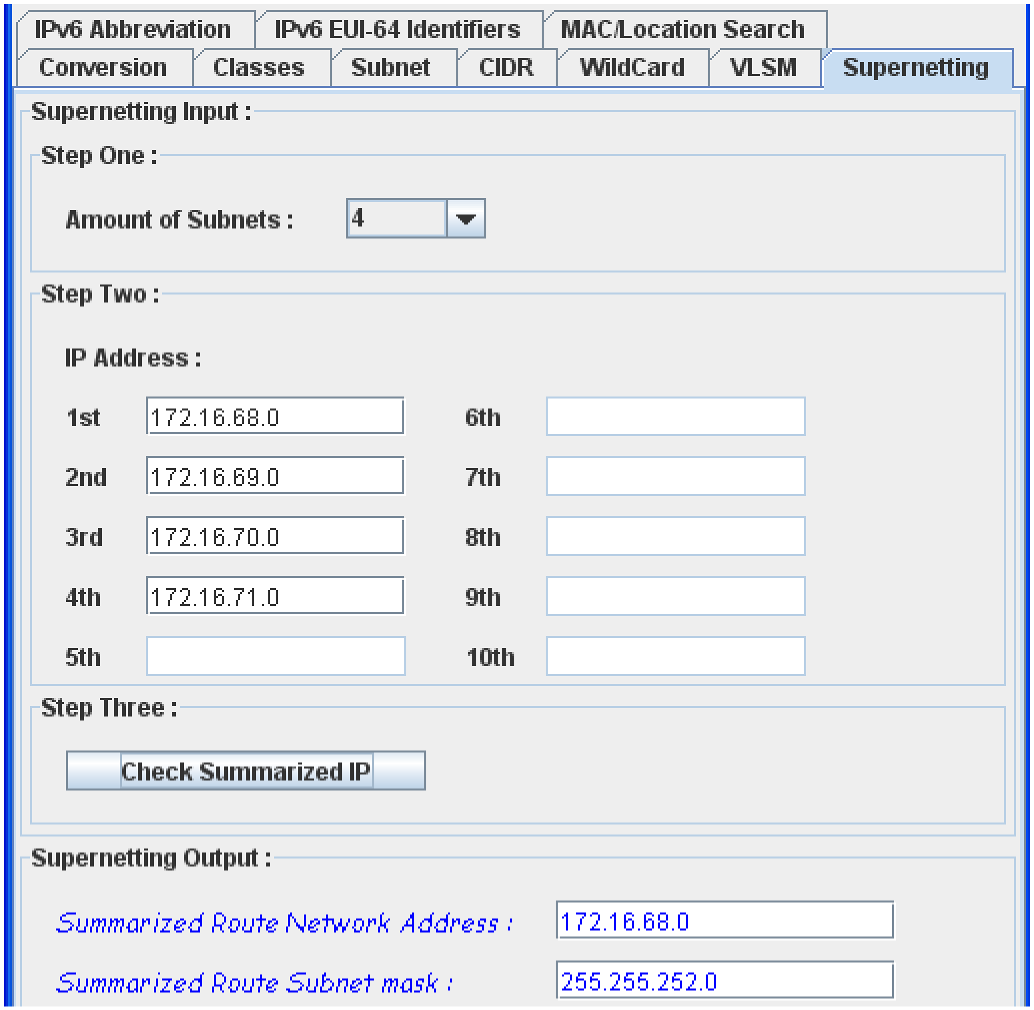
For Calgary, you do the same thing that you did for Winnipeg—look for common bit patterns in the routes:

172.16.68.0 = *10101100.00010000.010001*00.00000000
172.16.69.0 = *10101100.00010000.010001*01.00000000
172.16.70.0 = *10101100.00010000.010001*10.00000000
172.16.71.0 = *10101100.00010000.010001*11.00000000
Common bits: *10101100.00010000.010001*xx

Once again, the first 22 bits are common. The summarized route is therefore

172.16.68.0/22

Using UTAR IP Calculator



Route Summarization Example (4)

Step 3: Summarize Edmonton's Routes

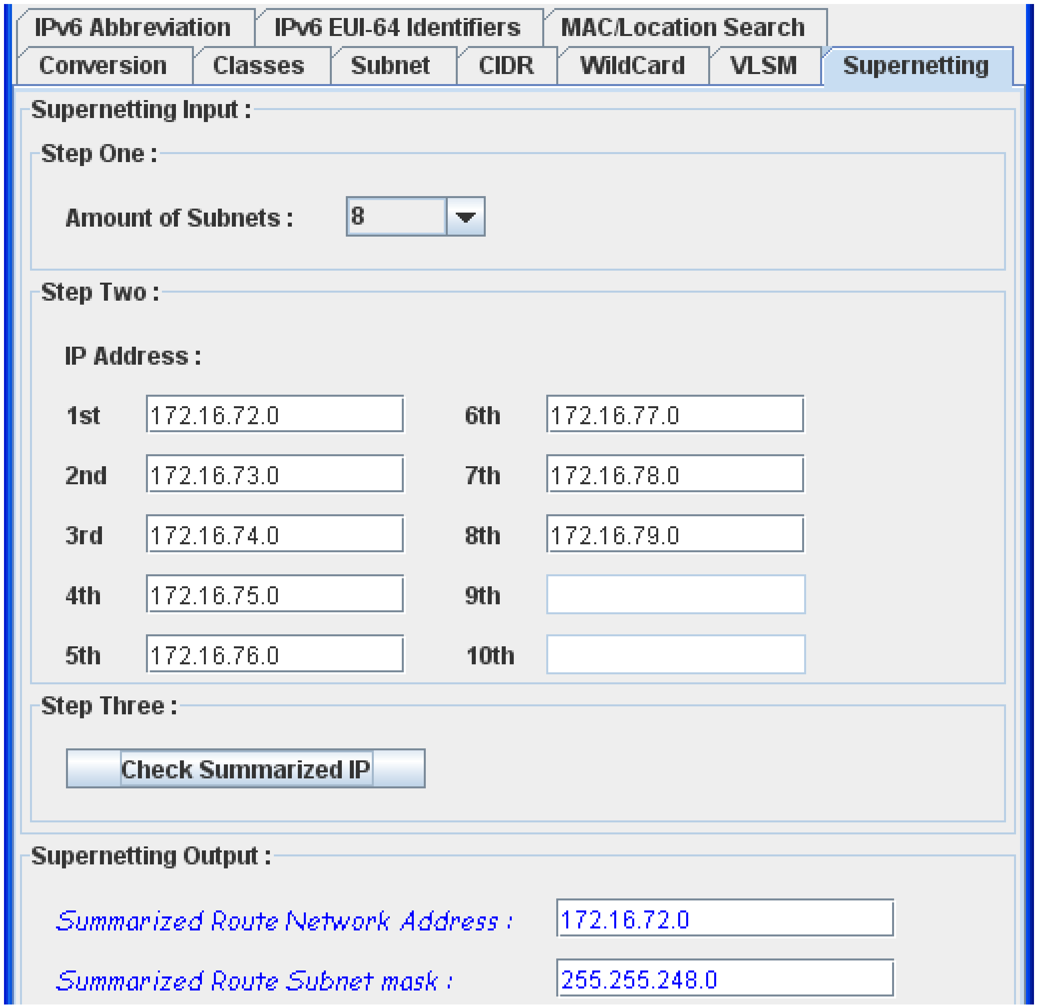
For Edmonton, you do the same thing that we did for Winnipeg and Calgary—look for common bit patterns in the routes:

172.16.72.0 = *10101100.00010000.01001*000.00000000
172.16.73.0 = *10101100.00010000.01001*001.00000000
172.16.74.0 = *10101100.00010000 01001*010.00000000
172.16.75.0 = *10101100.00010000 01001*011.00000000
172.16.76.0 = *10101100.00010000.01001*100.00000000
172.16.77.0 = *10101100.00010000.01001*101.00000000
172.16.78.0 = *10101100.00010000.01001*110.00000000
172.16.79.0 = *10101100.00010000.01001*111.00000000
Common bits: *10101100.00010000.01001*xxx

For Edmonton, the first 21 bits are common. The summarized route is therefore

172.16.72.0/21

Using UTAR IP Calc



Route Summarization Example (5)

Step 4: Summarize Vancouver's Routes

Yes, you can summarize Vancouver's routes to Seattle. You continue in the same format as before. Take the routes that Winnipeg, Calgary, and Edmonton sent to Vancouver, and look for common bit patterns:

172.16.64.0 = *10101100.00010000.0100*0000.00000000
172.16.68.0 = *10101100.00010000.0100*0100.00000000
172.16.72.0 = *10101100.00010000.0100*1000.00000000
Common bits: *10101100.00010000.0100*xxxx

Because there are 20 bits that are common, you can create one summary route for Vancouver to send to Seattle:

172.16.64.0/20

Try this supernetting with UTAR IP calculator (yourself)

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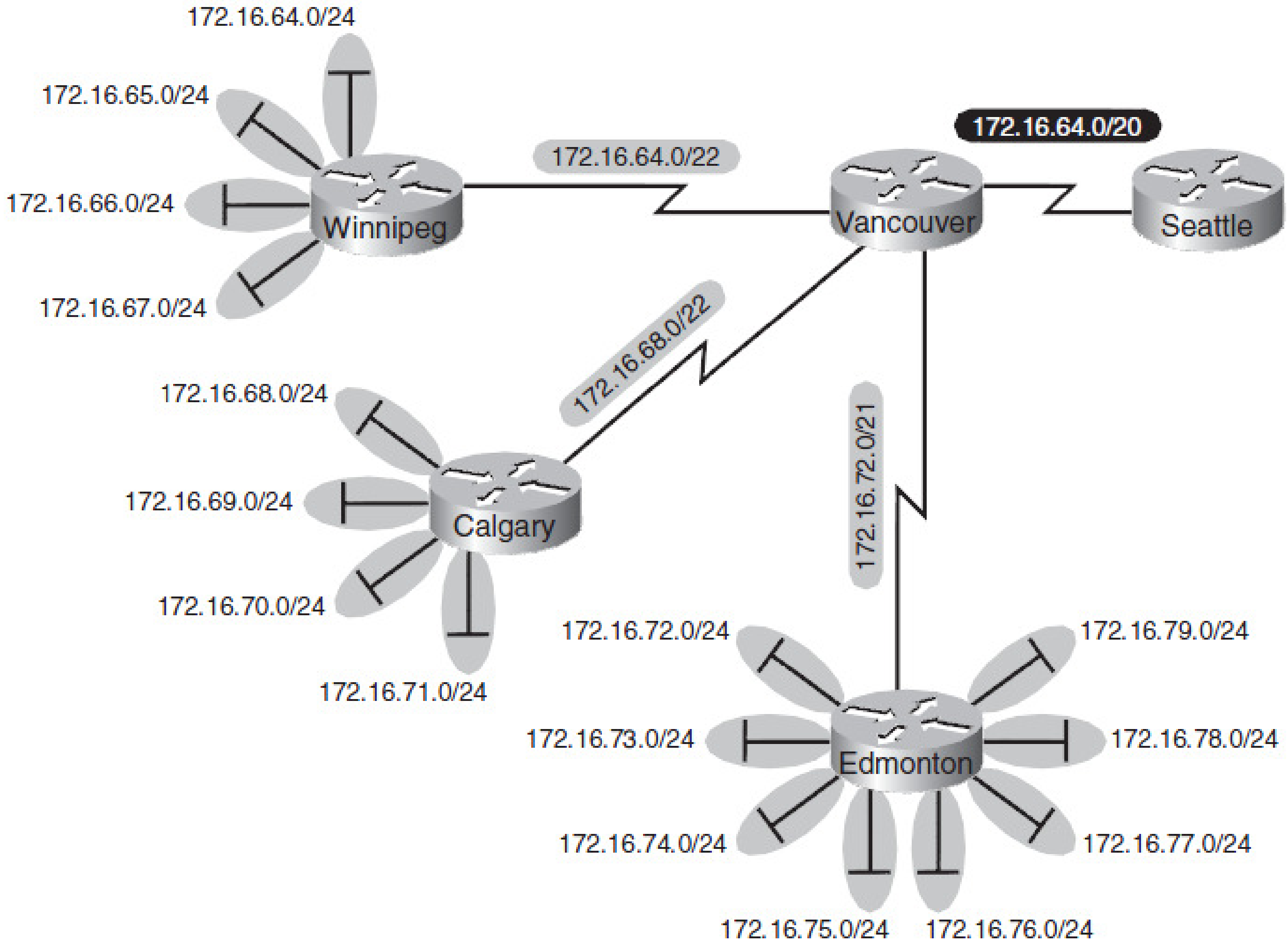
Route Summarization Example (6)

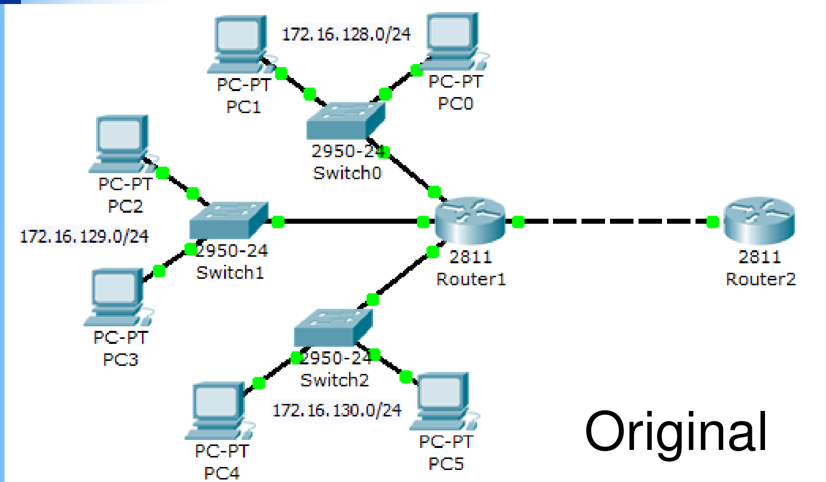


Supernetting & Route Summarization

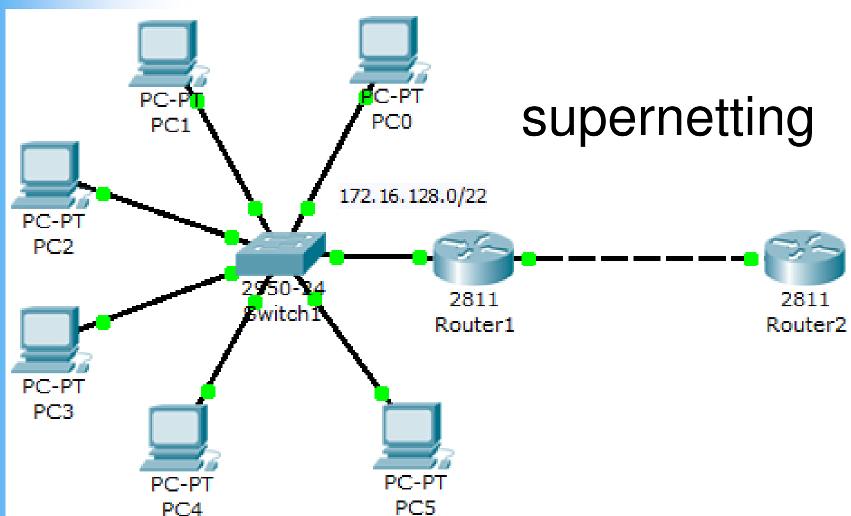
- Supernet and route summarization use the same IP address computation that combines a few IP of a few subnets into one.

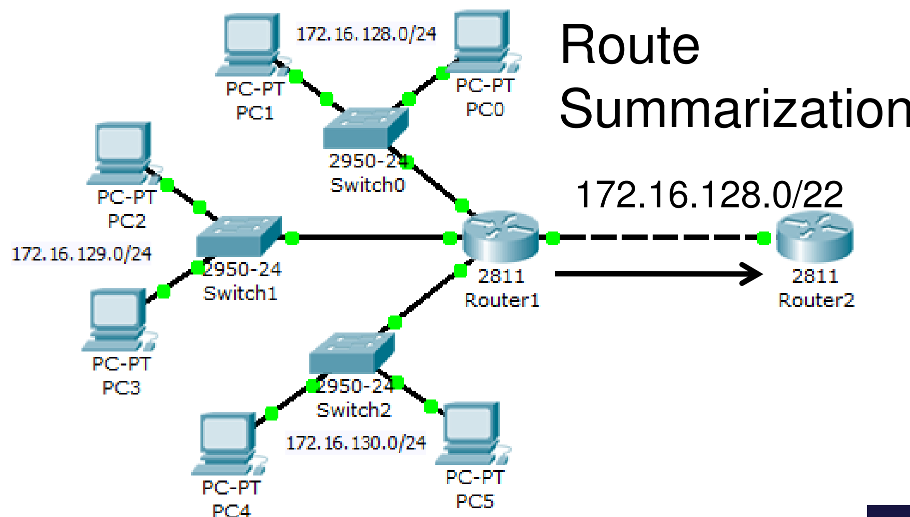
- The difference is that supernetting will physically combine a few subnets together whereas route summarization only compute the supernet network ID (in router) without changing the network.

Supernetting & Route Summarization



- Both route summarization & supernetting produce the same summarized IP network ID.
  - However supernetting changes the network topology.





Cisco Route Summarization

- Cisco routers can summarize routes in two ways:
  - Automatically, by summarizing subprefixes to the classful network boundary when crossing classful network boundaries (automatic summary).
  - As specifically configured (or manually), advertising a summarized local IP address pool on the specified interface